

# Homework 1

## Problem 1

(a)

$$\begin{aligned}\varepsilon \dot{v} &= (1 - v^2)v - w \\ \dot{w} &= av - w\end{aligned}$$

We can determine the equations for the isoclines:

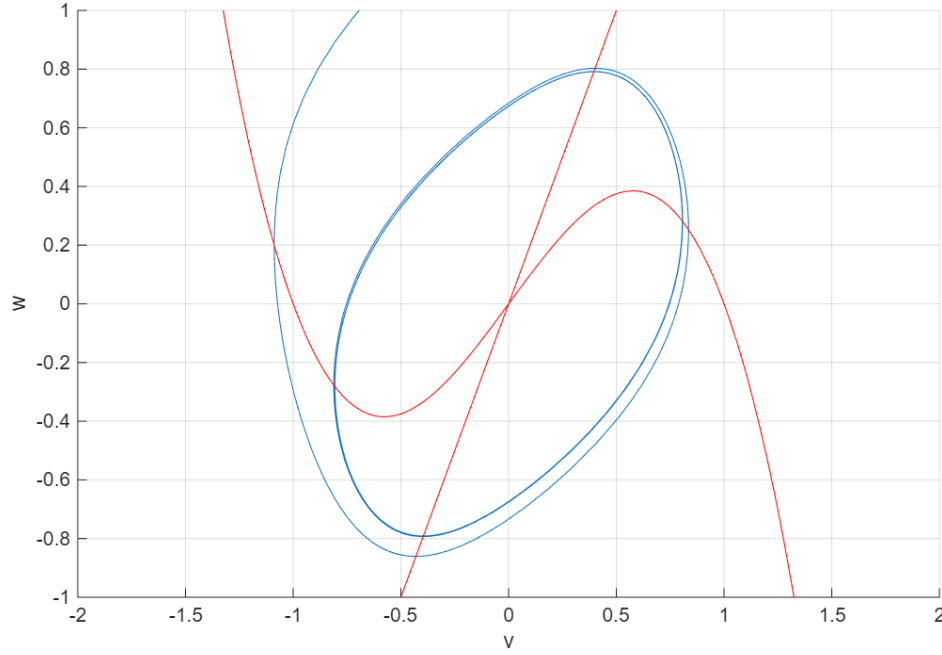
$$\begin{aligned}\dot{v} &= 0 \\ \implies w &= (1 - v^2)v\end{aligned}$$

$$\begin{aligned}\dot{w} &= 0 \\ \implies w &= av\end{aligned}$$

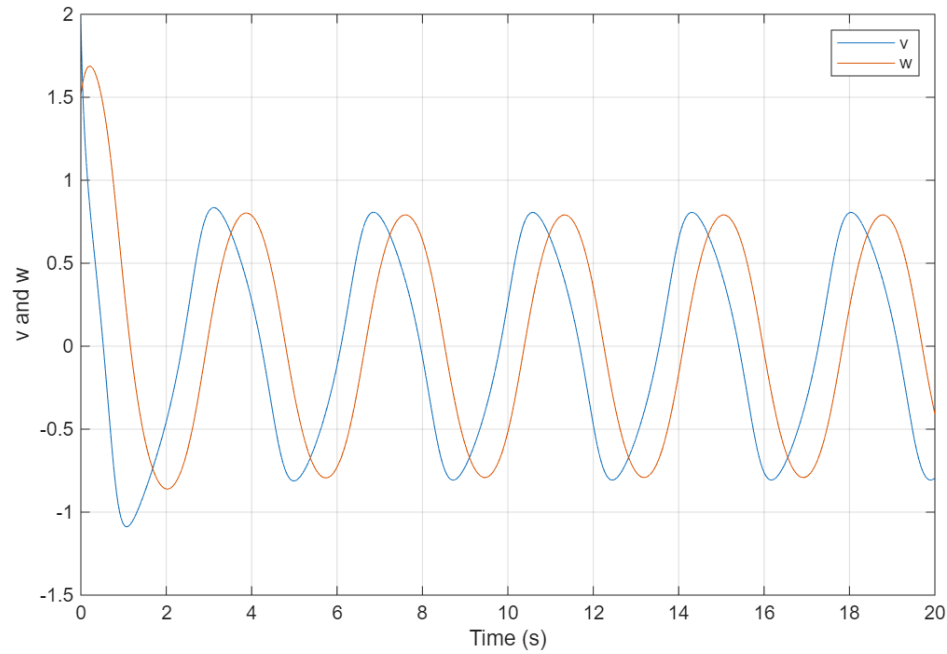
It would be useful to obtain the dynamic system into the form  $\dot{x} = f(x)$ , as it will more appropriate for the simulations we will run.

$$\underbrace{\begin{bmatrix} \dot{v} \\ \dot{w} \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} \frac{1-v^2}{\varepsilon} & -\frac{1}{\varepsilon} \\ a & -1 \end{bmatrix}}_{f(x)} \underbrace{\begin{bmatrix} v \\ w \end{bmatrix}}_{x}$$

We set  $\varepsilon = 0.5$  and  $a = 2$ . Choosing, an initial state  $x_0 = \begin{bmatrix} 2 & 1.5 \end{bmatrix}^T$ , we can find the solution to this system using MATLAB's `ode45()` function.



**Figure 1:** State trajectory of dynamical system with  $\varepsilon = 0.5$  and  $a = 2$ . Initial condition  $x_0 = \begin{bmatrix} 2 & 1.5 \end{bmatrix}^T$ .



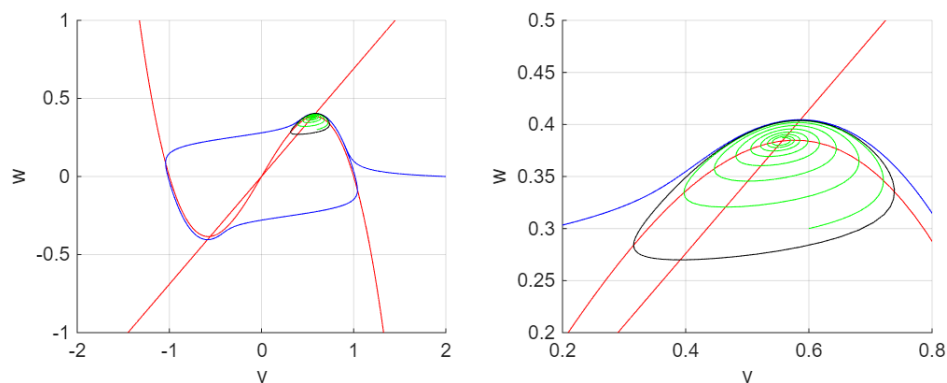
**Figure 2:** Solution to dynamical system with  $\varepsilon = 0.5$  and  $a = 2$ . Initial condition  $x_0 = \begin{bmatrix} 2 & 1.5 \end{bmatrix}^T$ .

We see that the solution to the dynamical system is periodic from Figure 2. Each cycle of the solution corresponds to a single orbit in the state space trajectory in Figure 1. We see that the transient behavior is captured by both figure, indicated by the pre-periodic behavior exhibited in both plots. Every crossing of an isocline in Figure 1 corresponds to a point in the solution where  $\dot{v}$  or  $\dot{w}$  is zero. In other words, at each crossing of an isocline, one of the states in Figure 2 is at its crest or trough of its path, where  $v(t)$  peaks

at the right intersection,  $w(t)$  peaks at the top intersection,  $v(t)$  bottoms at the left intersection, and  $w(t)$  bottoms at the bottom intersection.

(b)

We alter the values of the dynamical system to be  $\varepsilon = 0.1$  and  $a = 0.69$ . We can now use this system and plot the trajectories of solutions for different initial conditions.



**Figure 3:** State trajectory of solutions to the dynamical system. Right image is a magnified portion of the left image. Black line represents the domain of attraction of one of the stable equilibria.

This is cool. We expect that any solutions with an initial condition within the black line will converge to its respective stable equilibrium. Any other initial condition will converge to either the other stable equilibrium or will converge to the periodic orbit denoted by the blue line.

The location of the stable equilibrium is also quite intuitive and makes sense. It lies at the intersection of the two isoclines, which means that at that state both  $\dot{v} = \dot{w} = 0$ . In another representation,  $\dot{x} = 0$  and the state does not change, indicating that it is a stable equilibrium point. We can extrapolate from this behavior, where we can expect similar behavior about the opposing intersection point of the two isoclines. Perhaps this could be a necessary condition for stability, as it would seem that the origin is not a stable equilibrium point and would converge to the stable limit cycle.

## Problem 2

(a)

Something that we need is the  $M$  matrix from the equation  $u = Mv$ .

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -\sigma v_2 \\ -\sigma v_3 \\ -\sigma v_1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -\sigma & 0 \\ 0 & 0 & -\sigma \\ -\sigma & 0 & 0 \end{bmatrix}}_M \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

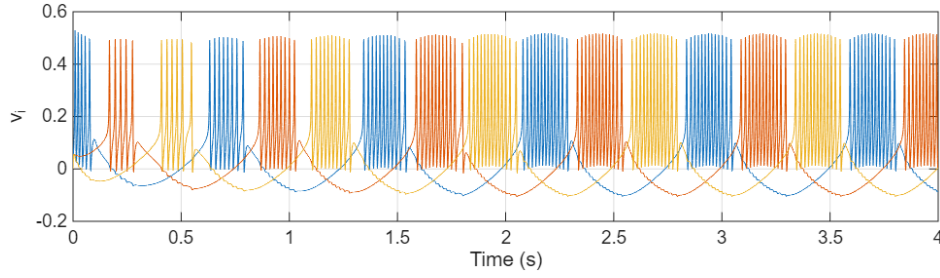
We can represent the equation from  $u$  to  $q$  into the state space.

$$q = f(s)u \iff \begin{aligned} \dot{x} &= Ax + Bu \\ q &= Cx \end{aligned}$$

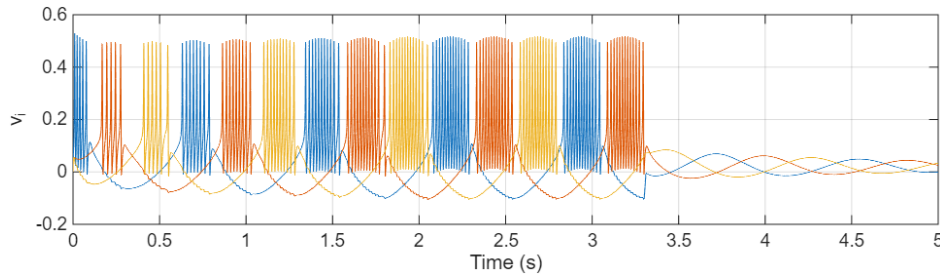
where  $x$  is some internal dynamic state. We can use this alongside our **axon** model

$$\begin{aligned} c\dot{v}_i &= \psi(v_i) - w_i + q_i \\ \tau\dot{w}_i &= \phi(v_i) - w_i \end{aligned}$$

We can use these equations to simulate the membrane potential  $v$  with  $\sigma = 2$ .



**Figure 4:** Oscillation of RCI with dynamic axon. Pulse input of magnitude 2.2 applied to  $u_1$  from  $t \in [0 \ 0.01]$  seconds.



**Figure 5:** Oscillation of RCI with dynamic axon turned off. Additional pulse input of magnitude 3.5 applied to  $u_1$  from  $t \in [3.3 \ 3.31]$  seconds.

Even though the RCI oscillation can be turned off, the oscillation is still **stable**. The initial pulse pushes the dynamic state into the domain of attraction of the limit cycle. The oscillations seen in Figures 4 and 5 represent the trajectory of a stable periodic orbit in the state space of the system. If left unperturbed, it would be safe to assume that the system will remain in this periodic orbit. What the "off" input pulse does is knock the state space trajectory out of this limit cycle and into the domain of attraction of a stable equilibrium point.

(b)

The MHB equation is:

$$\begin{aligned} \hat{q} &= f(j\omega)MK_\alpha\hat{q}, \quad \hat{q}_i = \alpha_i e^{j\beta_i} \\ \hat{q} &= \frac{-\sigma\kappa(a)}{1+j\omega\tau_0} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \hat{q} \end{aligned}$$

For the solution with equal amplitude oscillations,  $K_\alpha$  will become

$$K_\alpha = \kappa(a)I$$

where

$$\kappa(a) = \frac{2}{\pi a} \int_0^\pi \tanh(a \sin \theta) \sin \theta \, d\theta$$

So, we can work and mold the MHB equation:

$$\hat{q} = \frac{-\sigma\kappa(a)}{1+j\omega\tau_0} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \hat{q}$$

$$-\frac{1+j\omega\tau_0}{\sigma\kappa(a)} \hat{q} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \hat{q}$$

The fraction on the left is a scalar. Thus, this is functionally an eigenvalue/eigenvector equation. Thus, we can equate the fraction to different eigenvalues of the right side's matrix to determine different values for the parameters.

$$0 = \det \left( \lambda I - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \right)$$

$$= \det \left( \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -1 & 0 & \lambda \end{bmatrix} \right)$$

$$= \lambda^3 - 1$$

$$\lambda^3 = 1$$

$$\lambda = e^{j2\pi k/3}, \quad k = 0, 1, 2$$

We can solve for parameters for each eigenvalue. For  $k = 0$ :

$$-\frac{1+j\omega\tau_0}{\sigma\kappa(a)} = 1$$

$$\sigma\kappa(a) = -1 - j\omega\tau_0$$

Imaginary:  $-\omega\tau_0 = 0$

$$\omega = 0$$

Because we want the solution for oscillatory behavior, we can remove this as a candidate mode. Moving on to  $k = 1$ :

$$-\frac{1+j\omega\tau_0}{\sigma\kappa(a)} = e^{j2\pi/3}$$

$$= -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$

$$1+j\omega\tau_0 = \frac{\sigma\kappa(a)}{2} - j\frac{\sigma\kappa(a)\sqrt{3}}{2}$$

Real:  $1 = \frac{\sigma\kappa(a)}{2}$

$$\kappa(a) = \frac{2}{\sigma}$$

Imaginary:  $\omega\tau_0 = -\frac{\sigma\kappa(a)\sqrt{3}}{2}$

$$\omega = -\frac{\sqrt{3}}{\tau_0}$$

Let's pause here. Moving on to  $k = 2$ :

$$\begin{aligned}
 -\frac{1 + j\omega\tau_0}{\sigma\kappa(a)} &= e^{j4\pi/3} \\
 &= -\frac{1}{2} - j\frac{\sqrt{3}}{2} \\
 1 + j\omega\tau_0 &= \frac{\sigma\kappa(a)}{2} + j\frac{\sigma\kappa(a)\sqrt{3}}{2} \\
 \text{Real: } 1 &= \frac{\sigma\kappa(a)}{2} \\
 \kappa(a) &= \frac{2}{\sigma} \\
 \text{Imaginary: } \omega\tau_0 &= \frac{\sigma\kappa(a)\sqrt{3}}{2} \\
 \omega &= \frac{\sqrt{3}}{\tau_0}
 \end{aligned}$$

It appears that we obtain the same expression for  $\kappa(a)$  for either eigenvalue. The only difference lies in the sign of the angular frequency of the oscillation. When this is realized, this simply affects the phase at which the sinusoid oscillates. Both should be valid, but for ease of calculation, we will be using the case where  $k = 2$ .

Moving on to phase calculation, we know that that information will be embedded into  $\hat{q}$ . We know that this must be the eigenvector associated with the eigenvalue of interest.

$$\begin{aligned}
 \left( \lambda I - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \right) \hat{q} &= 0 \\
 \begin{bmatrix} e^{j4\pi/3} & -1 & 0 \\ 0 & e^{j4\pi/3} & -1 \\ -1 & 0 & e^{j4\pi/3} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} &= 0 \\
 q_2 &= q_1 e^{j4\pi/3} \\
 q_3 &= q_2 e^{j4\pi/3} = q_1 e^{j2\pi/3}
 \end{aligned}$$

The choice of  $q_1$  is somewhat arbitrary. As such, we will choose it to be  $e^{j0} = 1$ , and the following elements will just be phase-shifted from this value.

$$\hat{q} = \begin{bmatrix} 1 \\ e^{j4\pi/3} \\ e^{j2\pi/3} \end{bmatrix}$$

We can extract the phase information from this (at least relative to the first element of  $\hat{q}$ ).

$$\begin{aligned}
 \omega &= \frac{\sqrt{3}}{\tau_0} \\
 a &= \kappa^{-1} \left( \frac{2}{\sigma} \right) \\
 \beta_1 &= 0 \\
 \beta_2 &= 240^\circ \\
 \beta_3 &= 120^\circ
 \end{aligned}$$

This solution is expected to exist so long as  $0 < \kappa(a) < 1$ . We can use this to define a valid range for  $\sigma$ :

$$\boxed{\sigma > 2}$$

Orbital stability of the oscillation can be evaluated by looking at the eigenvalues of the QLS.

$$\begin{aligned} q &= f(s)MK_\alpha q \\ 0 &= (I - f(s)MK_\alpha)q \end{aligned}$$

The eigenvalues of the QLS are the values of  $s$  that make  $\det(I - f(s)MK_\alpha) = 0$ . We need to ensure that the real components of these eigenvalues are less than or equal to zero when we apply our found values of  $(\omega, a)$ .

$$\begin{aligned} \det(I - f(s)MK_\alpha) &= \det \left( \begin{bmatrix} 1 & \frac{\sigma\kappa(a)}{1+\tau_0 s} & 0 \\ 0 & 1 & \frac{\sigma\kappa(a)}{1+\tau_0 s} \\ \frac{\sigma\kappa(a)}{1+\tau_0 s} & 0 & 1 \end{bmatrix} \right) \\ 0 &= 1 + \left( \frac{\sigma\kappa(a)}{1 + \tau_0 s} \right)^3 \\ -\frac{\sigma\kappa(a)}{1 + \tau_0 s} &= \sqrt[3]{1} = e^{j2\pi k/3}, \quad k = 0, 1, 2 \end{aligned}$$

Case  $k = 0$ :

$$\begin{aligned} -\frac{\sigma\kappa(a)}{1 + \tau_0 s} &= 1 \\ 1 + \tau_0 s &= -\sigma\kappa(a) \\ &= -2 \\ s &= -\frac{3}{\tau_0} \end{aligned}$$

Case  $k = 1$ :

$$\begin{aligned} -\frac{\sigma\kappa(a)}{1 + \tau_0 s} &= e^{j2\pi/3} \\ &= -\frac{1}{2} + j\frac{\sqrt{3}}{2} \\ 1 + \tau_0 s &= -\frac{2\sigma\kappa(a)}{-1 + j\sqrt{3}} \\ &= -\frac{4 \cdot (-1 - j\sqrt{3})}{4} \\ &= 1 + j\sqrt{3} \\ s &= j\frac{\sqrt{3}}{\tau_0} \end{aligned}$$

Case  $k = 2$ :

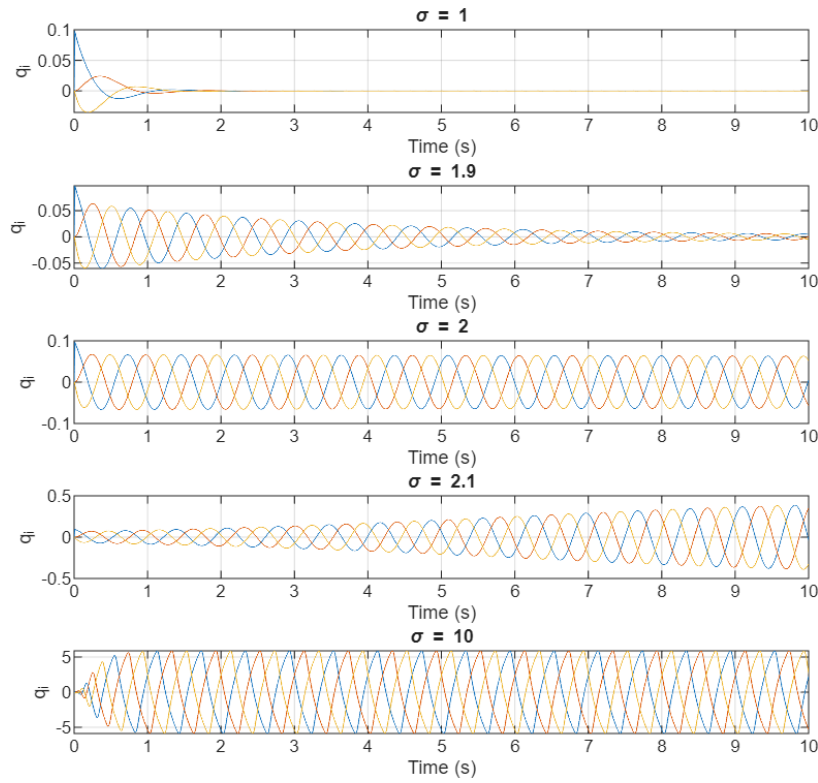
$$\begin{aligned}
-\frac{\sigma\kappa(a)}{1+\tau_0s} &= e^{j4\pi/3} \\
&= -\frac{1}{2} - j\frac{\sqrt{3}}{2} \\
1+\tau_0s &= -\frac{2\sigma\kappa(a)}{-1-j\sqrt{3}} \\
&= -\frac{4 \cdot (-1+j\sqrt{3})}{4} \\
&= 1-j\sqrt{3} \\
s &= -j\frac{\sqrt{3}}{\tau_0}
\end{aligned}$$

Thus, we have obtained that the eigenvalues of the QLS are

$$\text{eig}(\text{QLS}) = \left\{ -\frac{3}{\tau_0}, j\frac{\sqrt{3}}{\tau_0}, -j\frac{\sqrt{3}}{\tau_0} \right\}$$

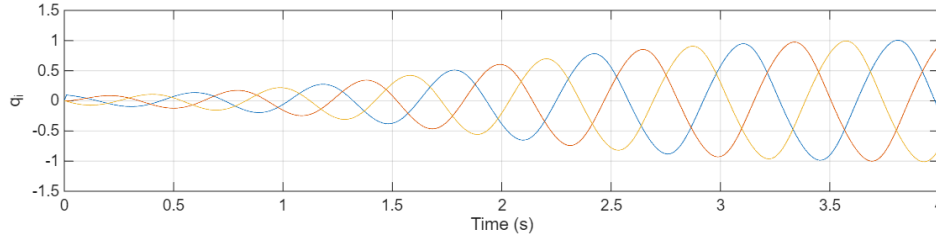
This validates that our solution is orbitally stable, as we have a pair of marginally stable eigenvalues.

(c)

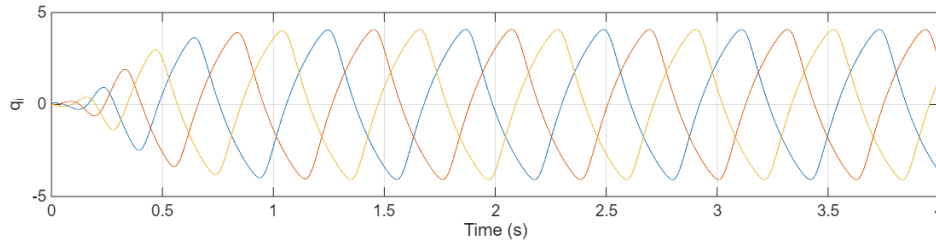


**Figure 6:** RCI behavior with varying values of  $\sigma$ .

The results from Figure 6 verify our range for  $\sigma$ . For values of  $\sigma$  less than 2, we see that oscillatory behavior starts, but eventually decay to a zero state. This indicates that the state trajectory remains in the domain of attraction of the stable zero-equilibrium point. At values of 2 and higher, we see that  $q_i$  settle into oscillatory motion. This indicates that the oscillation is orbitally stable, which verifies our range of  $\sigma$ .



**Figure 7:** Oscillation of RCI with static axon ( $\sigma = 2.5$ ).



**Figure 8:** Oscillation of RCI with static axon ( $\sigma = 7$ ).

|                |       | Predicted |       |                     | Empirical |       |                     |
|----------------|-------|-----------|-------|---------------------|-----------|-------|---------------------|
|                |       | $\omega$  | $a_i$ | $\beta_i - \beta_1$ | $\omega$  | $a_i$ | $\beta_i - \beta_1$ |
| $\sigma = 2.5$ | $q_1$ | 8.66      | 1.041 | $0^\circ$           | 8.74      | 1.027 | $0^\circ$           |
|                | $q_2$ | 8.66      | 1.041 | $240^\circ$         | 8.74      | 1.027 | $-120^\circ$        |
|                | $q_3$ | 8.66      | 1.041 | $120^\circ$         | 8.74      | 1.027 | $-240^\circ$        |
| $\sigma = 7$   | $q_1$ | 8.66      | 4.354 | $0^\circ$           | 10.12     | 4.080 | $0^\circ$           |
|                | $q_2$ | 8.66      | 4.354 | $240^\circ$         | 10.12     | 4.080 | $-120^\circ$        |
|                | $q_3$ | 8.66      | 4.354 | $120^\circ$         | 10.12     | 4.079 | $-240^\circ$        |

**Table 1:** MHB predicted and simulated empirical parameters of  $q_i$ .

It appears that the accuracy of the MHB predictions depend on the size of  $\sigma$ . For the small value case ( $\sigma = 2.5$ ), the prediction lies very close to the empirical results from the simulation. The difference between  $(\omega, a)$  are very similar to another. In regards to the phase of the signals, they are actually the same, as the same once wrapped by  $360^\circ$ .

The larger value case ( $\sigma = 7$ ) proves itself to be less accurate. The simulations demonstrate a higher frequency and lower amplitude oscillations. Moreover, the actual signals appear less like a single sinusoidal function, as shown in Figure 8. This aligns with some expectations we have. At such large values of  $\sigma$  and large amplitude oscillations, the linear model we use to approximate  $v_i$  less accurate

to the shape of  $\tanh(q_i)$ . Thus, as oscillations become more extreme, we expect more deviation from prediction to simulation.

## Homework 2

### Problem 1

(a)

We know that the connectivity matrix  $M$  that satisfies the MHB equation and QLS marginal stability condition for the CPG satisfies:

$$MK(\alpha) = \begin{bmatrix} R & N \end{bmatrix} \begin{bmatrix} \Omega & X \\ 0 & Y \end{bmatrix} \begin{bmatrix} R^+ \\ N^+ \end{bmatrix}$$

We can start by rewriting our target oscillation to be positive-amplitude sine functions with phase shift.

$$q(t) \cong \begin{bmatrix} a \sin(\omega t + \frac{\pi}{2}) \\ a \sin(\omega t) \\ a \sin(\omega t + \pi) \end{bmatrix}$$

We glean some information from this.

$$K(\alpha) = \begin{bmatrix} \kappa(a) & 0 & 0 \\ 0 & \kappa(a) & 0 \\ 0 & 0 & \kappa(a) \end{bmatrix} = \kappa(a)I_3$$

$$\begin{aligned} R &= \begin{bmatrix} a \sin(\frac{\pi}{2}) & a \cos(\frac{\pi}{2}) \\ a \sin(0) & a \cos(0) \\ a \sin(\pi) & a \cos(\pi) \end{bmatrix} \\ &= \begin{bmatrix} a & 0 \\ 0 & a \\ 0 & -a \end{bmatrix} \end{aligned}$$

$$\begin{aligned} N &\triangleq \text{null}(R^T) \\ &= \text{null}\left(\begin{bmatrix} a & 0 & 0 \\ 0 & a & -a \end{bmatrix}\right) \\ &= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
R^+ &= (R^T R)^{-1} R^T \\
&= \begin{bmatrix} a^2 & 0 \\ 0 & 2a^2 \end{bmatrix}^{-1} \begin{bmatrix} a & 0 & 0 \\ 0 & a & -a \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{a^2} & 0 \\ 0 & \frac{1}{2a^2} \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & a & -a \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{a} & 0 & 0 \\ 0 & \frac{1}{2a} & -\frac{1}{2a} \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
N^+ &= (N^T N)^{-1} N^T \\
&= [2]^{-1} \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}
\end{aligned}$$

$$\Omega = \begin{bmatrix} 1 & -\bar{\omega} \\ \bar{\omega} & 1 \end{bmatrix}$$

Letting  $X \in \mathbb{R}^{2 \times 1}$  and  $Y \in \mathbb{R}$

$$\begin{aligned}
MK(\alpha) &= \begin{bmatrix} R & N \end{bmatrix} \begin{bmatrix} \Omega & X \\ 0 & Y \end{bmatrix} \begin{bmatrix} R^+ \\ N^+ \end{bmatrix} \\
\kappa(a)M &= \begin{bmatrix} R\Omega & RX + NY \end{bmatrix} \begin{bmatrix} R^+ \\ N^+ \end{bmatrix} \\
&= \begin{bmatrix} a & -a\bar{\omega} & ax_1 \\ a\bar{\omega} & a & ax_2 + y \\ -a\bar{\omega} & -a & -ax_2 + y \end{bmatrix} \begin{bmatrix} \frac{1}{a} & 0 & 0 \\ 0 & \frac{1}{2a} & -\frac{1}{2a} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \\
&= \begin{bmatrix} 1 & \frac{ax_1 - \bar{\omega}}{2} & \frac{ax_1 + \bar{\omega}}{2} \\ \bar{\omega} & \frac{y + 1 + ax_2}{2} & \frac{y - a + x_2}{2} \\ -\bar{\omega} & \frac{y - a - ax_2}{2} & \frac{y + 1 - x_2}{2} \end{bmatrix} \\
M &= \frac{1}{2\kappa(a)} \begin{bmatrix} 2 & ax_1 - \bar{\omega} & ax_1 + \bar{\omega} \\ 2\bar{\omega} & y + 1 + ax_2 & y - 1 + ax_2 \\ -2\bar{\omega} & y - 1 - ax_2 & y + 1 - ax_2 \end{bmatrix}
\end{aligned}$$

Because  $x_1$  and  $x_2$  are arbitrary real scalars, we can incorporate the scalar constant  $a$  into them at no cost. Thus, we have that any satisfactory  $M$  is parameterized by:

$$M = \frac{1}{2\kappa(a)} \begin{bmatrix} 2 & x_1 - \bar{\omega} & x_1 + \bar{\omega} \\ 2\bar{\omega} & y + 1 + x_2 & y - 1 + x_2 \\ -2\bar{\omega} & y - 1 - x_2 & y + 1 - x_2 \end{bmatrix}$$

(b)

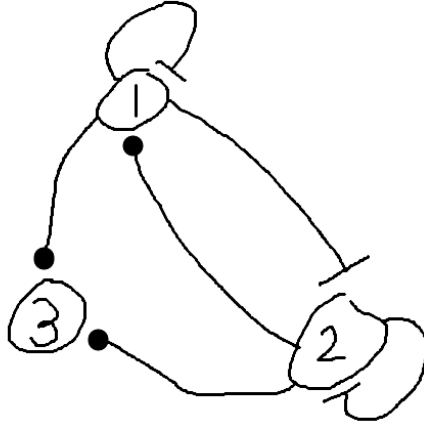
The prescribed behavior is achieved when the connectivity matrix  $M$  has a third column of just zeros. Thus, a satisfactory choice in parameters is:

$$(x_1, x_2, y) = (-\bar{\omega}, 1, 0)$$

The resulting connectivity matrix is:

$$M = \frac{1}{\kappa(a)} \begin{bmatrix} 1 & -\bar{\omega} & 0 \\ \bar{\omega} & 1 & 0 \\ -\bar{\omega} & -1 & 0 \end{bmatrix}$$

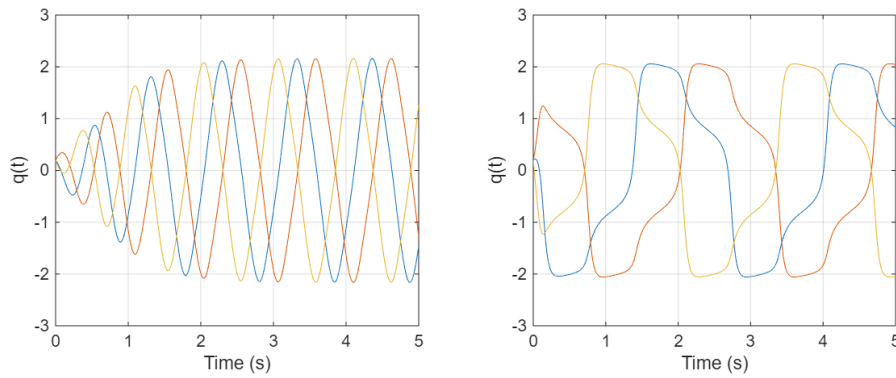
The neural network connections are shown in the following figure:



**Figure 1:** Neural network diagram.

(c)

Setting amplitude  $a = 2$ , frequency  $\omega = 2\pi$ , and initial condition  $q(0) = \text{col}(1, 1, 1)/5$ , we can simulate the system with varying low-pass filter cut-off frequencies.



**Figure 2:** Simulated three-neuron CPG with  $w_0 = \pi$  (left) and  $w_0 = 8\pi$  (right).

Letting these CPGs simulate for a duration of 200 seconds, we can determine their oscillation profiles for their steady state behavior.

|                   | $\omega$                             | $a_1$  | $a_2$  | $a_3$  | $\phi_1$          | $\phi_2$  | $\phi_3$          |
|-------------------|--------------------------------------|--------|--------|--------|-------------------|-----------|-------------------|
| $\omega_0 = \pi$  | $6.0577 \frac{\text{rad}}{\text{s}}$ | 2.0499 | 2.0459 | 2.0495 | $90.0239^\circ$   | $0^\circ$ | $180.0000^\circ$  |
| $\omega_0 = 8\pi$ | $2.3813 \frac{\text{rad}}{\text{s}}$ | 2.0260 | 2.0261 | 2.0261 | $-270.0016^\circ$ | $0^\circ$ | $-180.0000^\circ$ |

**Table 1:** Oscillation profile data for  $\omega_0 = \pi$  and  $\omega_0 = 8\pi$ 

These results are pretty neat. The first thing to note is that the phases of both oscillations meet the target phase values. Moreover, the amplitudes are sufficiently close to the target value, with only 2.5% and 1.3% error for each of the values of  $\omega_0$ .

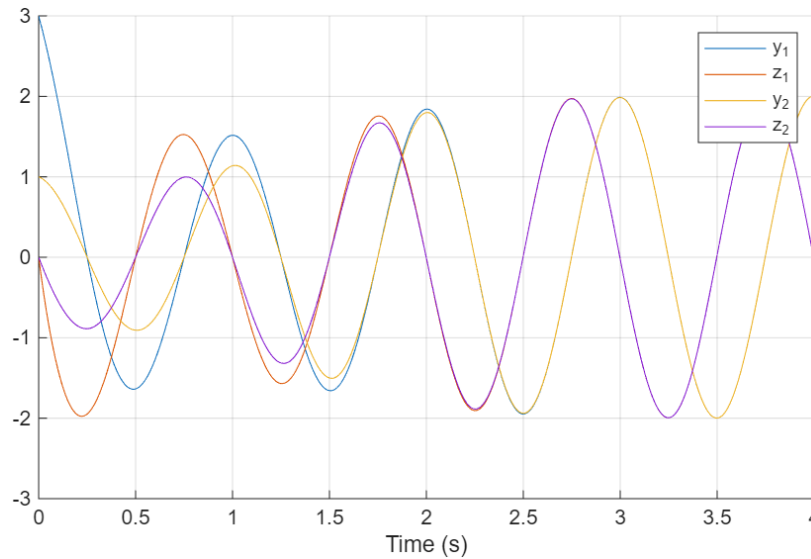
The main difference is between the fundamental frequency and the oscillation shape of the two trials. We can see that the  $\omega_0 = \pi$  more accurately reflects the target oscillation, with a fundamental frequency near the target  $\omega = 2\pi$  and visually looks more like a single-frequency sinusoid. On the other hand, the  $\omega_0 = 8\pi$  CPG is quite inaccurate to the target, with a large difference between the fundamental and target frequencies and the shape of the oscillation being very different from a perfect sinusoid.

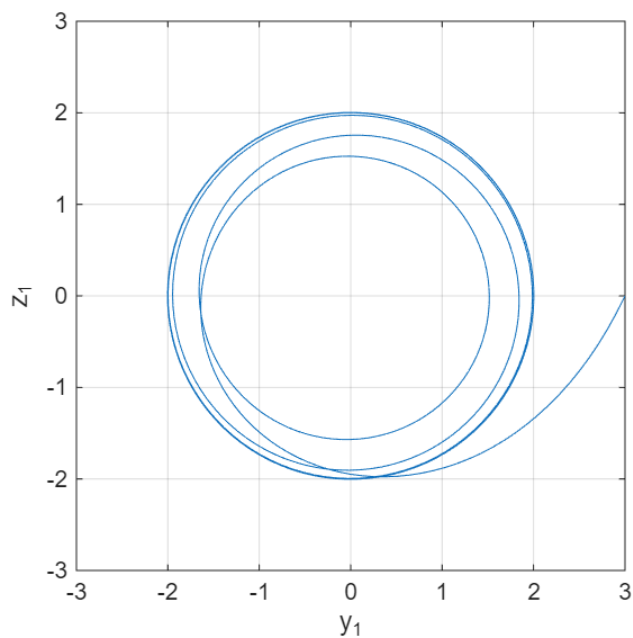
We can trace this back to the implications of changing  $\omega_0$ .  $\omega_0$  represents the cutoff frequency of the low pass filter  $f(s)$ . A low value of  $\omega_0$  means that only low frequency components of the CPG are retained in the system dynamics. This is a good property to have, as our target oscillation is a simple sinusoid with only a fundamental frequency. As  $\omega_0$  is increased, higher frequency components of the CPG are allowed into the system. This higher-frequency components have drastically altered the exhibited oscillation, altering the shape of the oscillation and frequency of oscillation.

## Problem 2

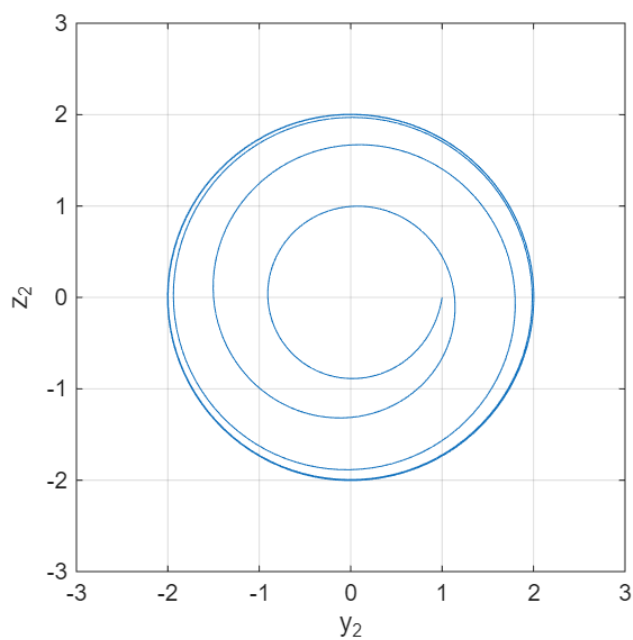
(a)

Simulating the two coupled AHOs, we obtain the following results.

**Figure 3:** Synchronized solution to coupled AHOs



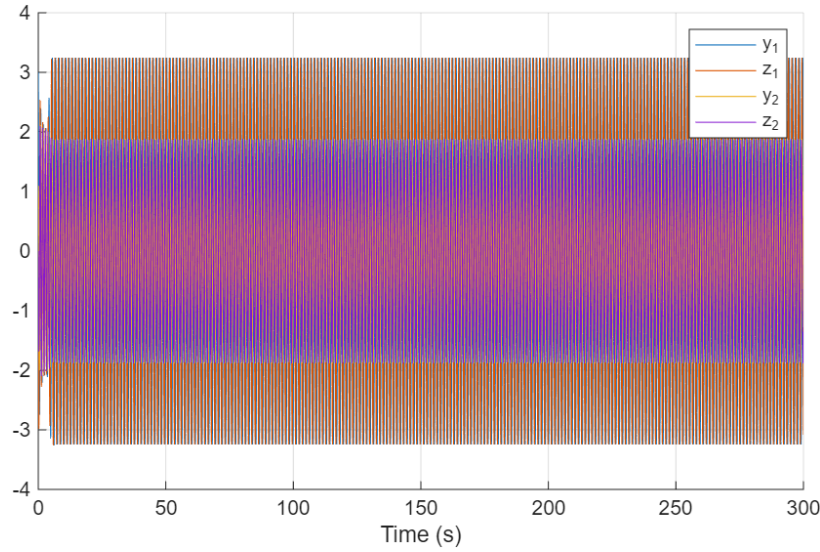
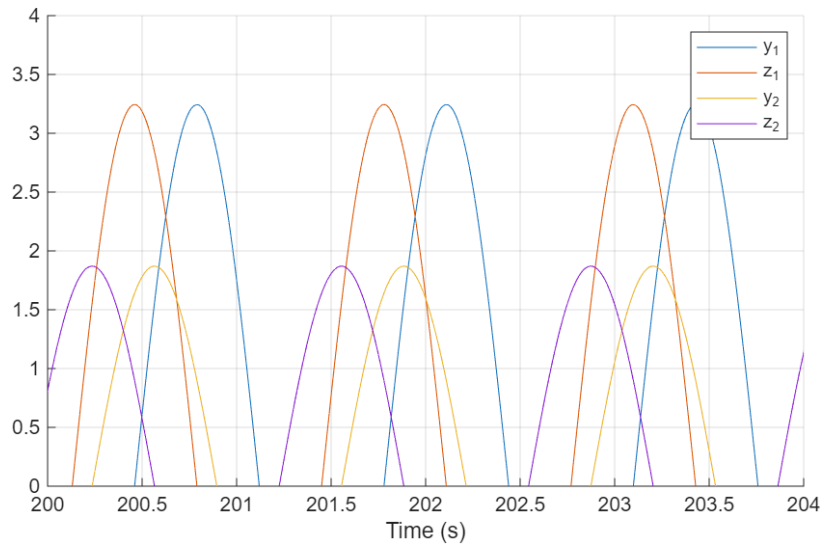
**Figure 4:** State trajectory of  $\chi_1$ .



**Figure 5:** State trajectory of  $\chi_2$ .

We see that the system achieves synchronization by the 2.5 second mark. Though not marked in the state trajectory plots, we would see that both oscillators enter the periodic orbit at the same state at the same time

(b)

**Figure 6:** Periodic solution to unsynchronized AHOs.**Figure 7:** Enlarged portion of unsynchronized solution.

I would argue that the synchronized periodic solution is not orbitally stable. The solution that the system settles into does not exhibit synchronization — the solutions  $\chi_1(t)$  and  $\chi_2(t)$  do not share phase or amplitude. It would appear that the orbital stability of the solution to the coupled AHOs is sensitive to the choice in  $\delta_1$  and  $\delta_2$ . However, that isn't to say that there isn't an orbitally stable solution to the system. We see clearly that the state trajectories fall into stable, but different, oscillations. This means that there is a stable limit cycle for the system, but it is not synchronized.

(c)

We begin by introducing the polar coordinates

$$\chi_1 = \begin{bmatrix} r_1 \sin \theta_1 \\ r_1 \cos \theta_1 \end{bmatrix}, \chi_2 = \begin{bmatrix} r_2 \sin \theta_2 \\ r_2 \cos \theta_2 \end{bmatrix}$$

Let's use these with our system dynamics. For an AHO  $\chi_i$ ,

$$\begin{aligned} \dot{\chi}_i &= \frac{d}{dt} \left( r_i \begin{bmatrix} \sin \theta_i \\ \cos \theta_i \end{bmatrix} \right) \\ &= \dot{r}_i \begin{bmatrix} \sin \theta_i \\ \cos \theta_i \end{bmatrix} + r_i \dot{\theta}_i \begin{bmatrix} \cos \theta_i \\ -\sin \theta_i \end{bmatrix} \\ &= \begin{bmatrix} c_i & s_i \\ -s_i & c_i \end{bmatrix} \begin{bmatrix} r_i \dot{\theta}_i \\ \dot{r}_i \end{bmatrix} \end{aligned}$$

Using the AHO dynamics, we can plug in our coordinate change.

$$\begin{aligned} \dot{\chi}_i &= \begin{bmatrix} \sigma(r_i) & \omega \\ -w & \sigma(r_i) \end{bmatrix} \begin{bmatrix} s_i \\ c_i \end{bmatrix} r_i + \delta_i \left( r_i \begin{bmatrix} s_i \\ c_i \end{bmatrix} + r_j \begin{bmatrix} s_j \\ c_j \end{bmatrix} \right) \\ &= \omega r_i \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} s_i \\ c_i \end{bmatrix} + \sigma(r_i) r_i \begin{bmatrix} s_i \\ c_i \end{bmatrix} + \delta_i r_i \begin{bmatrix} s_i \\ c_i \end{bmatrix} - \delta_i r_j \begin{bmatrix} s_j \\ c_j \end{bmatrix} \\ &= \omega r_i \begin{bmatrix} c_i \\ -s_i \end{bmatrix} + (\sigma(r_i) + \delta_i) r_i \begin{bmatrix} s_i \\ c_i \end{bmatrix} - \delta_i r_j \begin{bmatrix} s_j \\ c_j \end{bmatrix} \\ &= \begin{bmatrix} c_i & s_i \\ -s_i & c_i \end{bmatrix} \begin{bmatrix} \omega \\ (\sigma(r_i) + \delta_i) \end{bmatrix} r_i - \delta_i r_j \begin{bmatrix} s_j \\ c_j \end{bmatrix} \end{aligned}$$

In this formulation, the function  $\sigma$  represents  $\mu(\alpha^2 - \|z\|^2)$ . Because  $\theta_i$  has no impact on the magnitude of  $\|\chi_i\|$ , the function is represented as a function of  $r_i$  only. Equating the two equations we have derived:

$$\begin{aligned} \begin{bmatrix} c_i & s_i \\ -s_i & c_i \end{bmatrix} \begin{bmatrix} r_i \dot{\theta}_i \\ \dot{r}_i \end{bmatrix} &= \begin{bmatrix} c_i & s_i \\ -s_i & c_i \end{bmatrix} \begin{bmatrix} \omega \\ (\sigma(r_i) + \delta_i) \end{bmatrix} r_i - \delta_i r_j \begin{bmatrix} s_j \\ c_j \end{bmatrix} \\ \begin{bmatrix} r_i \dot{\theta}_i \\ \dot{r}_i \end{bmatrix} &= \begin{bmatrix} c_i & -s_i \\ s_i & c_i \end{bmatrix} \left( \begin{bmatrix} c_i & s_i \\ -s_i & c_i \end{bmatrix} \begin{bmatrix} \omega \\ (\sigma(r_i) + \delta_i) \end{bmatrix} r_i - \delta_i r_j \begin{bmatrix} s_j \\ c_j \end{bmatrix} \right) \\ &= \begin{bmatrix} \omega \\ (\sigma(r_i) + \delta_i) \end{bmatrix} r_i + \begin{bmatrix} s_i c_j - c_i s_j \\ -s_i s_j - c_i c_j \end{bmatrix} \delta_i r_j \\ \begin{bmatrix} r_i \dot{\theta}_i \\ \dot{r}_i \end{bmatrix} &= \begin{bmatrix} \omega \\ (\sigma(r_i) + \delta_i) \end{bmatrix} r_i + \begin{bmatrix} s_i - j \\ -c_i - j \end{bmatrix} \delta_i r_j \end{aligned}$$

We can define perturbation variables  $\rho_i$  and  $\phi_i$  and linearize with respect to them. We will assume that they are small and apply appropriate assumptions when necessary.

$$\rho := \begin{bmatrix} r_1 - \alpha \\ r_2 - \alpha \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}, \phi := \begin{bmatrix} \theta_1 - \omega t \\ \theta_2 - \omega t \end{bmatrix} = \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix}$$

$$\begin{aligned}
r_1 &= \rho_1 + \alpha \\
r_2 &= \rho_2 + \alpha \\
\theta_1 &= \phi_1 + \omega t \\
\theta_2 &= \phi_2 + \omega t \\
\dot{r}_1 &= \dot{\rho}_1 \\
\dot{r}_2 &= \dot{\rho}_2 \\
\dot{\theta}_1 &= \dot{\phi}_1 + \omega \\
\dot{\theta}_2 &= \dot{\phi}_2 + \omega
\end{aligned}$$

$$\begin{aligned}
\begin{bmatrix} (\rho_i + \alpha)(\dot{\phi}_i + \omega) \\ \dot{\rho}_i \end{bmatrix} &= \begin{bmatrix} \omega(\rho_i + \alpha) \\ \sigma(\rho_i + \alpha)(\rho_i + \alpha) + \delta_i(\rho_i + \alpha) \end{bmatrix} + \begin{bmatrix} \sin(\phi_i + \omega t - \phi_j - \omega t) \\ -\cos(\phi_i + \omega t - \phi_j - \omega t) \end{bmatrix} \delta_i(\rho_j + \alpha) \\
&= \begin{bmatrix} \omega(\rho_i + \alpha) \\ \sigma(\rho_i + \alpha)(\rho_i + \alpha) + \delta_i(\rho_i + \alpha) \end{bmatrix} + \begin{bmatrix} \sin(\phi_i - \phi_j) \\ -\cos(\phi_i - \phi_j) \end{bmatrix} \delta_i(\rho_j + \alpha)
\end{aligned}$$

The first simplification we can make is through the small angle assumption, where

$$\sin(\phi_i - \phi_j) \approx \phi_i - \phi_j, \cos(\phi_i - \phi_j) \approx 1$$

$$\Rightarrow \begin{bmatrix} (\rho_i + \alpha)(\dot{\phi}_i + \omega) \\ \dot{\rho}_i \end{bmatrix} = \begin{bmatrix} \omega(\rho_i + \alpha) \\ \sigma(\rho_i + \alpha)(\rho_i + \alpha) + \delta_i(\rho_i + \alpha) \end{bmatrix} + \begin{bmatrix} \phi_i - \phi_j \\ -1 \end{bmatrix} \delta_i(\rho_j + \alpha)$$

Our next focus can be the  $\sigma$  term.

$$\begin{aligned}
\sigma(\rho_i + \alpha)(\rho_i + \alpha) &= \mu(\alpha^2 - (\rho_i + \alpha)^2)(\rho_i + \alpha) \\
&= \mu(\alpha^2 - \rho_i^2 - 2\rho_i\alpha - \alpha^2)(\rho_i + \alpha)
\end{aligned}$$

The next simplification is through assuming that  $\rho_i$  is small. As such,  $\rho_i^2$  is even smaller and can be neglected. Moreover, we can make the assumption that  $\alpha \gg \rho_i$  and approximate  $\rho_i + \alpha \approx \alpha$ .

$$\begin{aligned}
\sigma(\rho_i + \alpha)(\rho_i + \alpha) &\approx \mu(-2\rho_i\alpha)\alpha \\
&= -2\mu\alpha^2\rho_i \\
&=: -\gamma\rho_i
\end{aligned}$$

Returning to our dynamics equation, we can substitute this value and also apply the small  $\rho$  assumption.

$$\begin{bmatrix} \alpha\dot{\phi}_i + \alpha\omega \\ \dot{\rho}_i \end{bmatrix} = \begin{bmatrix} \alpha\omega \\ -\gamma\rho_i + \delta_i(\rho_i + \alpha) \end{bmatrix} + \begin{bmatrix} \phi_i - \phi_j \\ -1 \end{bmatrix} \delta_i(\rho_j + \alpha)$$

We can work through this line-by-line.

$$\begin{aligned}
\alpha\dot{\phi}_i + \alpha\omega &= \alpha\omega + \delta_i(\rho_j + \alpha)(\phi_i - \phi_j) \\
\dot{\phi}_i &= \delta_i(\phi_i - \phi_j)
\end{aligned}$$

$$\begin{aligned}
\dot{\rho}_i &= -\gamma\rho_i + \delta_i(\rho_i + \alpha) - \delta_i(\rho_j + \alpha) \\
&= (\delta_i - \gamma)\rho_i - \delta_i\rho_j
\end{aligned}$$

We now have a pair of linear equations that describe the dynamics of the system. We can stack  $\rho_i$  equations and  $\phi_i$  equations, respectively.

$$\begin{aligned}\dot{\rho} &= \begin{bmatrix} \delta_1 - \gamma & -\delta_1 \\ -\delta_2 & \delta_2 - \gamma \end{bmatrix} \rho \\ \dot{\phi} &= \begin{bmatrix} \delta_1 & -\delta_1 \\ -\delta_2 & \delta_2 \end{bmatrix} \phi\end{aligned}$$

For synchronization, the matrix describing the dynamics of  $\phi$  has one eigenvalue at the origin and the other in the open left hand plane.

$$\begin{aligned}0 &= \det(\lambda I - N) \\ &= \det \left( \begin{bmatrix} \lambda - \delta_1 & \delta_1 \\ \delta_2 & \lambda - \delta_2 \end{bmatrix} \right) \\ &= (\lambda - \delta_1)(\lambda - \delta_2) - \delta_1 \delta_2 \\ &= \lambda^2 - (\delta_1 + \delta_2)\lambda \\ &= \lambda(\lambda - (\delta_1 + \delta_2))\end{aligned}$$

The above equation informs us that the eigenvalue at the origin is assured. The other eigenvalue is located at  $(\delta_1 + \delta_2)$ . Thus, our other condition is that

$$(\delta_1 + \delta_2) < 0$$

The next condition is that the eigenvalues of the matrix describing the  $\rho$  dynamics to have negative real parts.

$$\begin{aligned}0 &= \det(\lambda I - M) \\ &= \det \left( \begin{bmatrix} \lambda - (\delta_1 - \gamma) & \delta_1 \\ \delta_2 & \lambda - (\delta_2 - \gamma) \end{bmatrix} \right) \\ &= (\lambda - (\delta_1 - \gamma))(\lambda - (\delta_2 - \gamma)) - \delta_1 \delta_2 \\ &= \lambda^2 - (\delta_1 + \delta_2 - 2\gamma)\lambda + (\delta_1 - \gamma)(\delta_2 - \gamma) - \delta_1 \delta_2 \\ &= \lambda^2 - (\delta_1 + \delta_2 - 2\gamma)\lambda + \gamma(\gamma - (\delta_1 + \delta_2)) \\ \lambda &= \frac{\delta_1 + \delta_2 - 2\gamma \pm \sqrt{(\delta_1 + \delta_2 - 2\gamma)^2 - 4\gamma(\gamma - (\delta_1 + \delta_2))}}{2} \\ &= \frac{-2\gamma + (\delta_1 + \delta_2) \pm (\delta_1 + \delta_2)}{2} \\ \lambda &= -\gamma, (\delta_1 + \delta_2 - \gamma)\end{aligned}$$

By definition,  $\gamma = 2\mu\alpha^2$ . Because  $\alpha \in \mathbb{R}$ , we simply need to ensure that

$$\mu > 0$$

These two conditions also ensure that the other eigenvalue of  $M$  is negative, so they are sufficient in ensuring synchronization.

## Homework 3

The dynamics of the flipper locomoter are described by the following equations of motion:

$$\begin{aligned} J\ddot{\theta} + D\dot{\theta} + (\dot{w}\Lambda + k_o BB^T)\theta &= Bu \\ m\ddot{w} + d(\theta)\dot{w} + \dot{\theta}^T \Lambda \theta &= 0 \end{aligned}$$

Provided a target average velocity, we can formulate an optimization problem that minimizes the average input power.

$$\begin{aligned} \min_{u, \theta} & \frac{1}{T} \int_0^T \dot{\theta}^T B u \, dt \\ \text{s.t.} & \frac{1}{T} \int_0^T v(t) \, dt = v_o \end{aligned}$$

Through modeling the states as phasors, we can manipulate the the dynamics to obtain the following, equivalent optimization problem:

$$\begin{aligned} \min_{\omega, \hat{u}} & \hat{u}^* X_\omega \hat{u} \\ \text{s.t.} & \hat{u}^* Y_\omega \hat{u} = 1 \end{aligned}$$

where

$$\begin{aligned} X_\omega &= \frac{j\omega}{4} (B^T P_\omega - P_\omega^* B), \quad Y_\omega = P_\omega^* \left( \frac{j\omega S - v_o C_o}{2av_o} \right) P_\omega \\ P_\omega &= (J(j\omega)^2 + D(j\omega) + v_o \Lambda + k_o BB^T)^{-1} B, \quad S = \frac{\Lambda - \Lambda^T}{2} \end{aligned}$$

The minimization over two variables can be written as two minimizations over single variables.

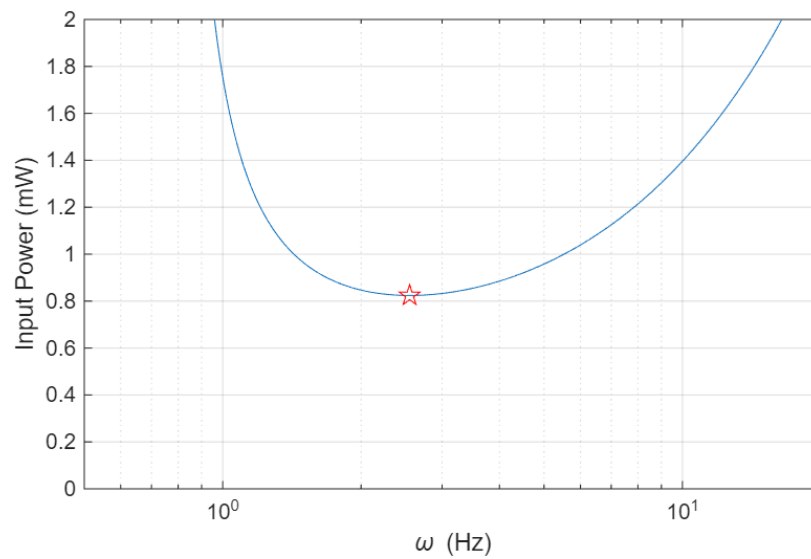
$$\min_{\omega} \min_{\hat{u}} (\hat{u}^* X_\omega \hat{u}) \quad \text{s.t.} \quad \hat{u}^* Y_\omega \hat{u} = 1$$

The minimization of  $\hat{u}$  lends itself well to the S-procedure technique, where the optimal cost/power will be the largest real generalized eigenvalue of  $(X_\omega, Y_\omega)$ ,  $\gamma_0$ , such that  $X_\omega \geq \gamma_0 Y_\omega$ . The optimal control input will be the eigenvector associated with  $\gamma_0$  normalized such that the constraint  $\hat{u}^* Y_\omega \hat{u} = 1$  is satisfied.

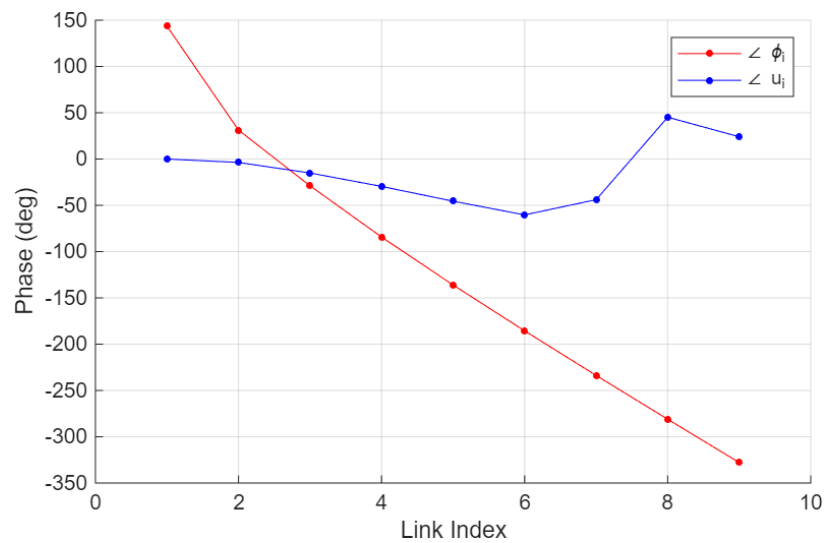
Based on this formulation, the optimal control input depends on the choice of  $\omega$ . For a range of frequencies, we can perform this optimization and obtain the associated power  $\gamma_\omega$ . We will then search for the optimal frequency  $\omega_o$  that has the minimum input power and use its optimal control input for our simulations. With our specified system parameters, we obtain the following optimal values:

$$\omega_o = 2.5441 \text{ Hz}$$

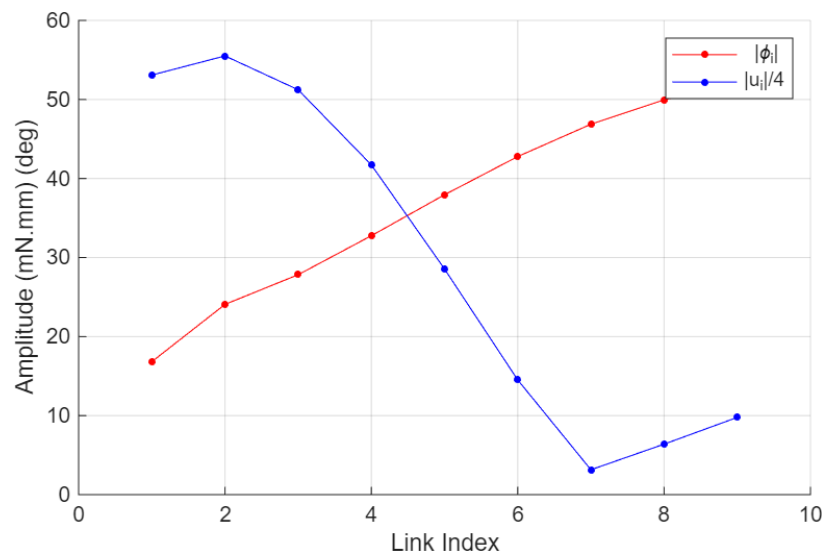
$$\gamma_{\min} = 0.8241 \text{ mW}$$



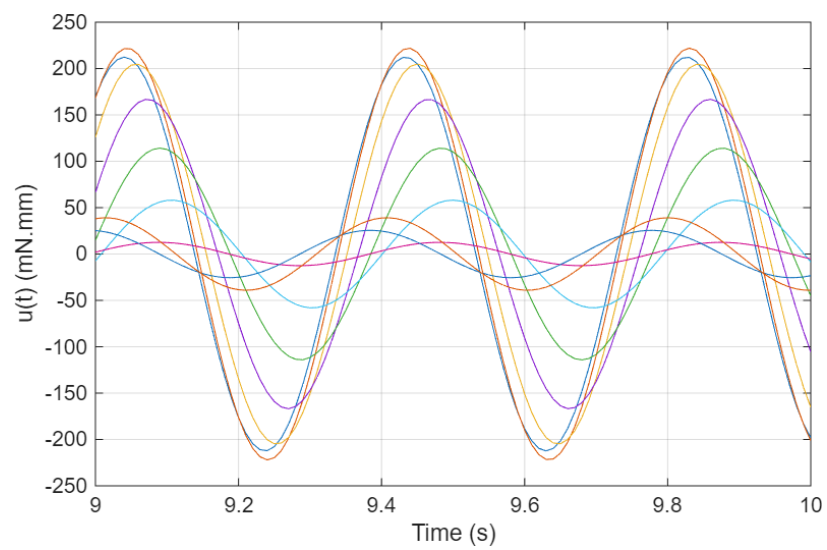
**Figure 1:** Minimum input power vs. frequency. Optimal frequency and associated power denoted by the red star.



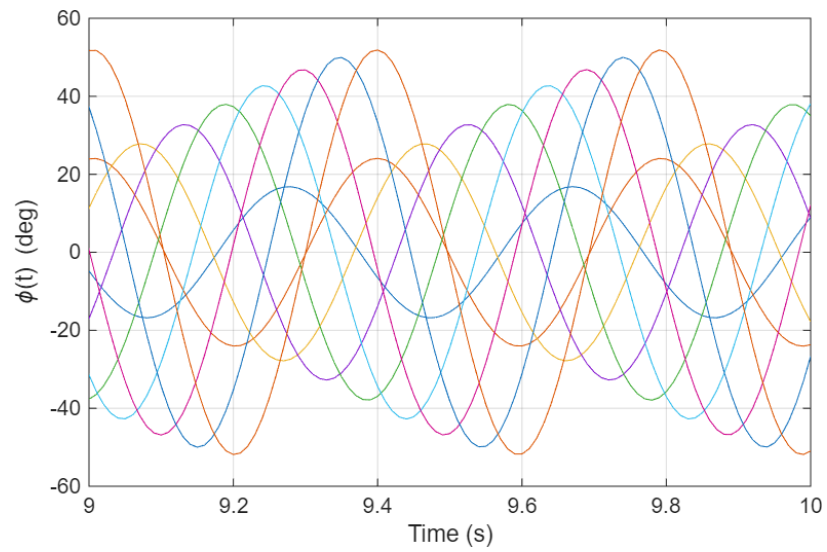
**Figure 2:** Phase of optimal gait and input, normalized such that  $\angle \hat{u}_1 = 0$ .



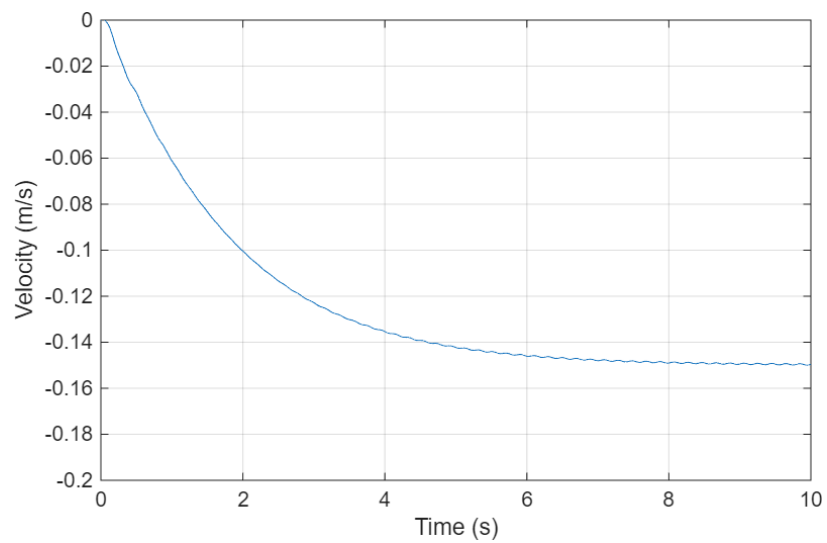
**Figure 3:** Amplitude of optimal gait and input.



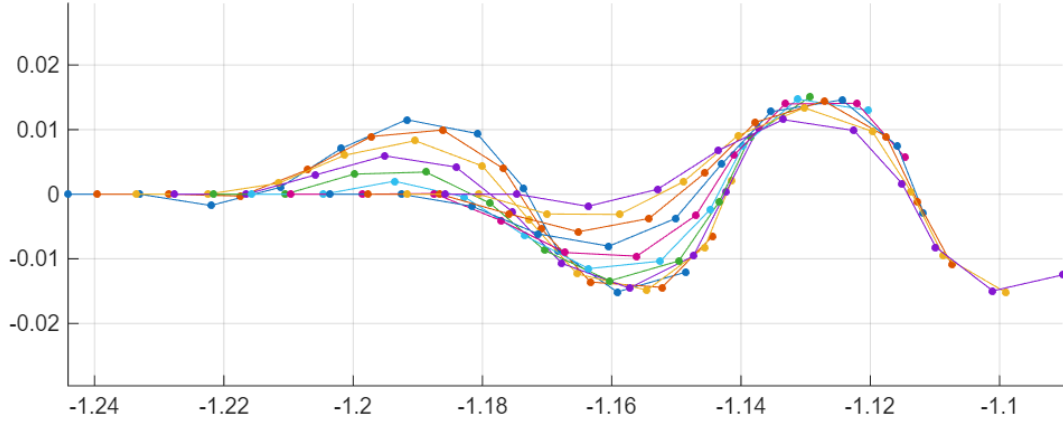
**Figure 4:** Optimal control input for  $9 \leq t \leq 10$ .



**Figure 5:** Simulated gait for  $9 \leq t \leq 10$ .



**Figure 6:** Simulated locomotor velocity.



**Figure 7:** Snapshots of locomotor from final period of oscillation.

These results are pretty nice. We can see in Figure 6 that the simulated locomotor eventually converges to the target velocity of -0.15 meters per second. Examining it closely, we see that there are actually very small oscillations occurring in the state, which is what we expect from our regularly oscillating gait. From Figure 2 we see that the optimal gait has equally spaced link phases, which is what we expected from a "flipper" with equal-length links. The phases span across 540 degrees, which we see reflected in the gait depicted in Figure 7. We can also notice that the amplitude of the gait increases as you move away from the head of the locomotor. Both of these properties are visualized through Figure 5, which depicts the simulated gait in the steady state.

In Figure 4 we see that the optimal control input does not have equal amplitude for each joint. Additionally, Figure 3 shows us that the further links have smaller amplitude, implying that the velocity gained from these links is not worth the power required to move these joints.

## Homework 4

(a)

We can start by getting the state dynamics down. We are given the linearized state dynamics, and we can use a basic integrator for the other two states:

$$\begin{aligned}\dot{\theta} &= \dot{\theta} \\ \ddot{\theta} &= a_1\theta + b_1u + \frac{1}{m\ell}d \\ \dot{x}_c &= \dot{x}_c \\ \ddot{x}_c &= a_2\theta + b_2u\end{aligned}$$

Now we truly have enough to formulate the generalized plant.

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{x}_c \\ \ddot{x}_c \\ \hbar_x x_c \\ u \\ y \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ a_1 & 0 & 0 & 0 & \frac{\hbar_d}{m\ell} & 0 & b_1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ a_2 & 0 & 0 & 0 & 0 & 0 & b_2 \\ 0 & 0 & \hbar_x & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \\ x_c \\ \dot{x}_c \\ d/\hbar_d \\ v \\ u \end{bmatrix}$$

Thus, the state space matrices are:

$$\begin{aligned}A &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ a_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ a_2 & 0 & 0 & 0 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0 & 0 \\ b_o & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ b_1 \\ 0 \\ b_2 \end{bmatrix} \\ C_1 &= \begin{bmatrix} 0 & 0 & \hbar_x & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \\ D_{11} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad D_{12} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad D_{21} = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D_{22} = \begin{bmatrix} 0 \end{bmatrix}\end{aligned}$$

(b)

Something we notice from the form of the dynamics is that we can get something in the form of:

$$\begin{bmatrix} \ddot{\theta} \\ \ddot{x}_c \end{bmatrix} + \begin{bmatrix} -a_1 & 0 \\ -a_2 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ x_c \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} u$$

We can denote a new state variable  $q$  as  $(\theta, x_c)$ . We can put this into phasor form:

$$\begin{aligned}
(j\omega)^2 \hat{q} + \begin{bmatrix} -a_1 & 0 \\ -a_2 & 0 \end{bmatrix} \hat{q} &= \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \hat{u} \\
\begin{bmatrix} (j\omega)^2 - a_1 & 0 \\ -a_2 & (j\omega)^2 \end{bmatrix} \hat{q} &= \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \hat{u} \\
\hat{q} &= \begin{bmatrix} (j\omega)^2 - a_1 & 0 \\ -a_2 & (j\omega)^2 \end{bmatrix}^{-1} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \hat{u} \\
&= \frac{1}{(j\omega)^2((j\omega)^2 - a_1)} \begin{bmatrix} (j\omega)^2 & 0 \\ a_2 & (j\omega)^2 - a_1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \hat{u} \\
&= \begin{bmatrix} \frac{1}{(j\omega)^2 - a_1} & 0 \\ \frac{a_2}{(j\omega)^2((j\omega)^2 - a_1)} & \frac{1}{(j\omega)^2} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \hat{u} \\
&= \begin{bmatrix} \frac{b_1}{(j\omega)^2 - a_1} \\ \frac{a_2 b_1 + b_2((j\omega)^2 - a_1)}{(j\omega)^2((j\omega)^2 - a_1)} \end{bmatrix} \hat{u}
\end{aligned}$$

Let's choose  $\hat{u} = ((j\omega)^2 - a_1)/b_1$ . We then get

$$\hat{q} = \begin{bmatrix} 1 \\ \frac{a_2 b_1 - b_2(\omega^2 + a_1)}{-b_1 \omega^2} \end{bmatrix}$$

Because  $\hat{q}$  is only encapsulating the position components, we multiply it by  $(j\omega)$  to get the velocity states. Composing the two, we can get the true state phasor,  $\hat{x}$ :

$$\hat{x} = \begin{bmatrix} 1 \\ j\omega \\ \frac{b_2(\omega^2 + a_1) - a_2 b_1}{b_1 \omega^2} \\ j \frac{b_2(\omega^2 + a_1) - a_2 b_1}{b_1 \omega} \end{bmatrix}$$

To get  $(X, U)$ , we can use  $P = \begin{bmatrix} \Im(\hat{p}) & \Re(\hat{p}) \end{bmatrix}$ .

$$\begin{aligned}
X &= \begin{bmatrix} 0 & 1 \\ \omega & 0 \\ 0 & \frac{b_2(\omega^2 + a_1) - a_2 b_1}{b_1 \omega^2} \\ \frac{b_2(\omega^2 + a_1) - a_2 b_1}{b_1 \omega} & 0 \end{bmatrix} \\
U &= \begin{bmatrix} 0 & -\frac{\omega^2 + a_1}{b_1} \end{bmatrix}
\end{aligned}$$

We need to check if these satisfy the desired initial condition properties. At time  $t = 0$ :

$$\begin{aligned}
x(0) &= Ae^{\Lambda \cdot 0} \eta \\
&= \begin{bmatrix} 0 & 1 \\ \omega & 0 \\ 0 & \frac{b_2(\omega^2 + a_1) - a_2 b_1}{b_1 \omega^2} \\ \frac{b_2(\omega^2 + a_1) - a_2 b_1}{b_1 \omega} & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 0 \\ \frac{b_2(\omega^2 + a_1) - a_2 b_1}{b_1 \omega^2} \\ 0 \end{bmatrix}
\end{aligned}$$

Thus, this condition is satisfied.

(c)

We know  $(K, F)$  to simply be the LQR and Kalman filter gains of the generalized plant. Thus, they can be obtained through solving for the stabilizing solutions to their respective Riccati equations.

$$\begin{aligned}
K &= -B_2^T P, \quad PA + A^T P - PB_2 B_2^T P + C_1^T C_1 = 0 \\
F &= -QC_2^T, \quad AQ + QA^T - QC_2^T C_2 Q + B_1 B_1^T = 0
\end{aligned}$$

Doing so for our numerical values, we get:

$$K = \begin{bmatrix} 216.3015 & 130.3231 & 10.0000 & 19.3029 \end{bmatrix}$$

$$F = \begin{bmatrix} 8.4853 \\ 16.6209 \\ -6.1364 \\ -18.8278 \end{bmatrix}$$

We obtain  $(Y, V)$  by solving the dual Regulator Equation. This is done by finding the minimizing arguments to the following optimization problem:

$$\begin{aligned}
\min_{Y, V} \quad & \text{tr}(TB_1 - XVD_{21})^T P (TB_1 - XVD_{21}) + \text{tr}(KT + UY)Q(KT + UY)^T \\
\text{s.t.} \quad & YA + VC_2 = \Lambda Y
\end{aligned}$$

where we use the matrices defined as before and  $T \triangleq I - XY$ . This appears to be very nasty to solve. However, we can notice that the constraint is affine in  $(Y, V)$  and that the cost is quadratic in  $(Y, V)$ . Thus, these matrices can be solved in a quadratic program-like fashion, which can be computed relatively easily through numerical methods.

Doing so, we obtain the following numerical results for the matrices:

$$Y = \begin{bmatrix} 3.0857 & 0.8232 & 4.7829 & 1.2759 \\ 1.6464 & -1.5429 & 2.5519 & -2.3914 \end{bmatrix}$$

$$V = \begin{bmatrix} -5.1038 \\ 9.5658 \end{bmatrix}$$

(d)

The convergence of the state estimate to the true state means that we can expect  $(C_2\hat{x} - y)$  to converge to zero in the steady state. As such,  $\xi$  follows the following dynamics equation:

$$\dot{\xi} = \Lambda\xi$$

We know the solution to this system to be in the form of  $\bar{\xi}(t) = e^{\Lambda t}\bar{\xi}_0$ . From this form, it is clear to us that our desired  $\eta$  should be the initial condition to this solution.

One of the first things we can pick off is rewriting the disturbance and sensor noise into  $w$ :

$$w(t) = \begin{bmatrix} d(t)/h_d \\ v(t) \end{bmatrix} = \begin{bmatrix} 1/h_d \\ 0 \end{bmatrix} \delta(t) =: w_o \delta(t)$$

I will derive the error dynamics of  $e = x - \hat{x}$  and rewrite the dynamics of  $\xi$  in terms of  $e$ .

$$\begin{aligned} e &= x - \hat{x} \\ \dot{e} &= \dot{x} - \dot{\hat{x}} \\ &= Ax + B_1w + B_2u - A\hat{x} - B_2u - F(C_2\hat{x} - y) \\ \dot{e} &= (A + FC_2)e + (B_1 + FD_{21})w \end{aligned}$$

$$\begin{aligned} \dot{\xi} &= \Lambda\xi + (YF - V)(C_2\hat{x} - y) \\ &= \Lambda\xi + (YF - V)(-C_2e - D_{21}w) \end{aligned}$$

Let's define a new state variable  $\nu := \xi + Ye$ . We can see that in the steady state,  $\bar{\nu} \rightarrow \bar{\xi}$ . If we find an initial condition for  $\nu$ , we will find the initial condition for  $\xi$  and, thus,  $\eta$ .

Let's find the dynamics for  $\nu$ .

$$\begin{aligned} \dot{\nu} &= \dot{\xi} + Y\dot{e} \\ &= \Lambda\xi - \cancel{YFC_2e} - \cancel{YFD_{21}w} + VC_2e + VD_{21}w + Y\cancel{Ae} + \cancel{YFC_2e} + YB_1w + \cancel{YFD_{21}w} \\ &= \Lambda\xi + (YA + VC_2)e + (YB_1 + VD_{21})w \end{aligned}$$

From the dual regulator equation  $YA + VC_2 = \Lambda Y$ , we can make a substitution:

$$\begin{aligned} \dot{\nu} &= \Lambda\xi + \Lambda Ye + (YB_1 + VD_{21})w \\ \dot{\nu} &= \Lambda\nu + (YB_1 + VD_{21})w \end{aligned}$$

Cool :)

We know from linear systems theory that the response to an impulse input from zero state is equivalent to the free response from an appropriate initial condition. We can see that the solution to the specified input  $w(t)$  will be:

$$\bar{\nu}(t) = e^{\Lambda t}(YB_1 + VD_{21})w_o$$

Since we know that  $\bar{\nu}(t) = \bar{\xi}(t)$ , it is clear that:

$$\eta = (YB_1 + VD_{21})w_o$$

Utilizing the structure of our specific problem,  $\eta$  for the symbolic parameters will be

$$\eta = Y \begin{bmatrix} 0 \\ \frac{1}{m\ell} \\ 0 \\ 0 \end{bmatrix}$$

(e)

For the NOR, we have the dynamics equation

$$\dot{\xi} = \Lambda\xi + \sigma(\|\xi\|)\xi$$

For this to have  $\bar{\xi}(t)$  as a solution, we need that nonlinear term to go to zero. Thus, we need:

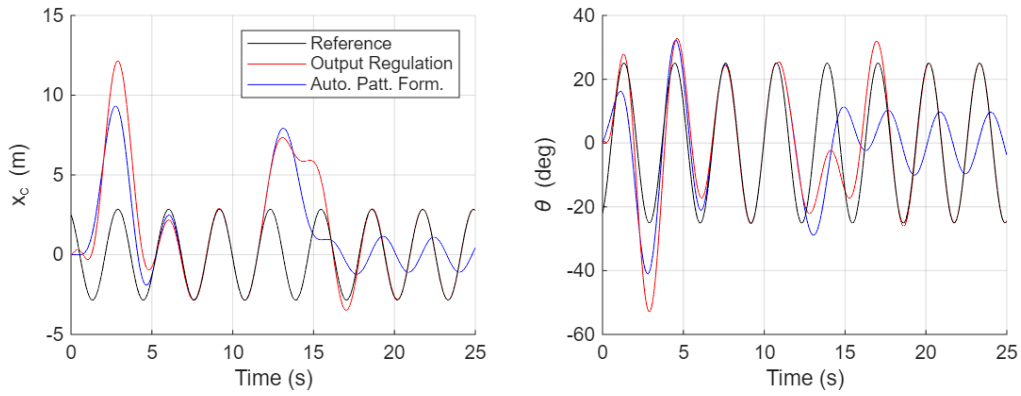
$$\mu(\alpha^2 - \|\bar{\xi}\|^2) = 0$$

We know that  $\bar{\xi}(t)$  is a simple sinusoid. Thus,  $\|\bar{\xi}\|$  will be constant throughout the trajectory. At  $t = 0$ ,  $\bar{\xi}(0) = \eta$ .  $\|\bar{\xi}\| = \|\eta\|$ . Thus,

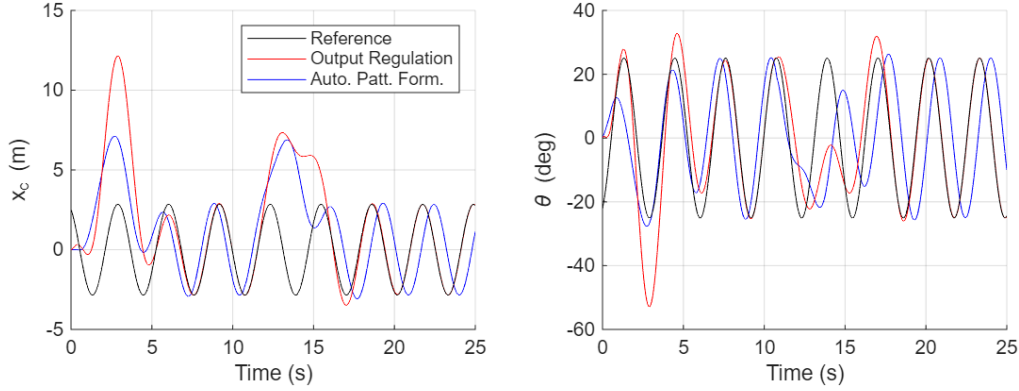
$$\alpha = \|\eta\|$$

(f)

We can simulate the closed-loop system under each of these controllers.



**Figure 1:** Closed-loop system under linear controllers. Target oscillations denoted in black.



**Figure 2:** Closed-loop system under nonlinear controllers. Target oscillations denoted in black.

That's neat. Something really interesting to see is that the output regulation controllers produced the same output regardless of linearity. This tracks with some intuition about these controllers. The output regulators have no consideration for the residuals in the oscillator dynamics. With no such slack term, these controllers will strictly track the system to the steady state response,  $x(t) \rightarrow X e^{\Lambda t} \eta$ . The nonlinearity's purpose is to maintain the same amplitude as the target signal. However, since the output regulators perfectly track to the target trajectory, their closed-loop behaviors will naturally have the same amplitude. Thus, there would be negligible difference between the linear and nonlinear controller, which is seen in the closed-loop responses being near-identical, shown in the Figures.

The cooler behavior is seen by the Autonomous Pattern Formation (APF) controller. Scoping outwards, we know that this choice in  $\psi(\xi)$  is to allow for a phase difference between the system and the target. We can recognize that the presence of the  $G$  term in the  $\xi$  dynamics allows for that slack to exist, thus changing the phase and amplitude of the system response. The effects of this is seen more clearly in the response of the LAPF controller. Though the phase difference may be hard to spot, the change in amplitude is very evident, with a large difference between the target and actualized oscillation amplitude. We correct for this with the selection of a nonlinear  $\psi(\xi)$  function. This specific choice pushes the amplitude to match that of the target trajectory. We see that this implementation is successful in Figure 2, where the simulated state trajectories have the same amplitude (but different phase) than the target trajectory. We see that there is *some* cost at the price of phase, where there is slightly more lag to the target in the nonlinear controller than in the linear controller. However, because amplitude is more pertinent to us, the NAPF is the better performing controller for our purposes.

## Final Exam

(a)

The dynamics of the flipper locomoter are described by the following equations of motion:

$$\begin{aligned} J\ddot{\theta} + D\dot{\theta} + (\dot{v}\Lambda + k_o BB^T)\theta &= Bu + F^T f \\ m\ddot{w} + d(\theta)\dot{w} + \dot{\theta}^T \Lambda \theta &= 0 \end{aligned}$$

For the sake of optimal gait derivation, we will assume the vertical disturbance force  $f = 0$ . Before we formulate the optimization problem, let us work our target cost into a more useful form.

$$\begin{aligned} \phi &= B^T \theta \\ \frac{d}{dt}(\phi) &= \frac{d}{dt}(B^T \theta) \\ \dot{\phi} &= B^T \dot{\theta} \end{aligned}$$

Thus, we can formulate our optimization problem:

$$\begin{aligned} \min_{\omega, \hat{u}} \quad & \frac{1}{T} \int_0^T \|B^T \dot{\theta}\|^2 dt \\ \text{s.t.} \quad & \frac{1}{T} \int_0^T v(t) dt = v_o \end{aligned}$$

Moving into phasor notation, we can manipulate the dynamics to obtain the following equivalent optimization problem:

$$\begin{aligned} \min_{\omega, \hat{u}} \quad & \hat{u}^* X_\omega \hat{u} \\ \text{s.t.} \quad & \hat{u}^* Y_\omega \hat{u} = 1 \end{aligned}$$

where

$$\begin{aligned} X_\omega &= \frac{\omega^2}{2} P_\omega^* B B^T P_\omega, \quad Y_\omega = P_\omega^* \left( \frac{j\omega S - v_o C_o}{2av_o} \right) P_\omega \\ P_\omega &= (J(j\omega)^2 + D(j\omega) + v_o \Lambda + k_o B B^T)^{-1} B, \quad S = \frac{\Lambda - \Lambda^T}{2} \end{aligned}$$

As we did in Lecture 12 and in Homework 3, we perform the minimization over two variables as a two minimizations over single variables:

$$\min_{\omega} \min_{\hat{u}} \hat{u}^* X_\omega \hat{u} \quad \text{s.t.} \quad \hat{u}^* Y_\omega \hat{u} = 1$$

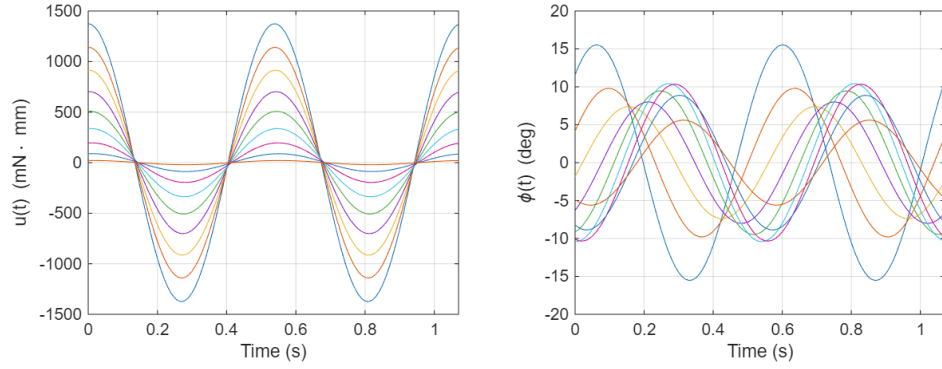
Again, we use the S-procedure where the optimal cost will be  $\gamma_0$ , the largest real generalized eigenvalue of  $(X_\omega, Y_\omega)$  such that  $X_\omega \geq \gamma_0 Y_\omega$ . The optimal control input will then be its associated eigenvector normalized such that  $\hat{u}^* Y_\omega \hat{u} = 1$ .

We then find the optimal value across a range of  $\omega$  and determine the minimizing choice of  $\omega$  and its associated control input  $\hat{u}$ .

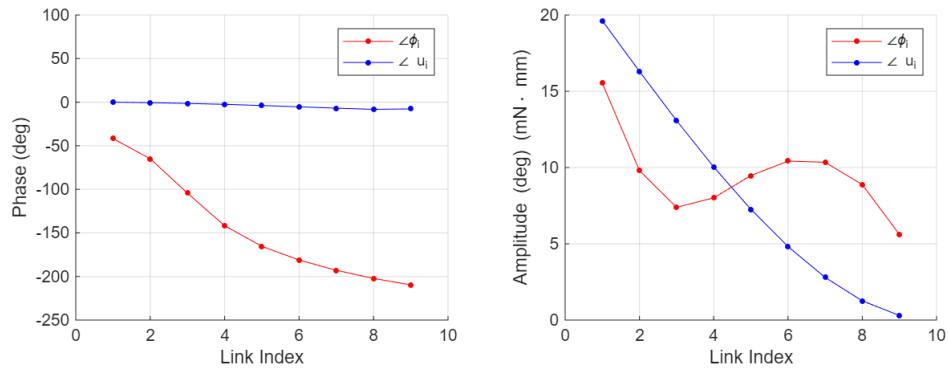
Performing this optimization, we obtain an optimal frequency:

$$\omega_o = 1.8563 \text{ Hz}$$

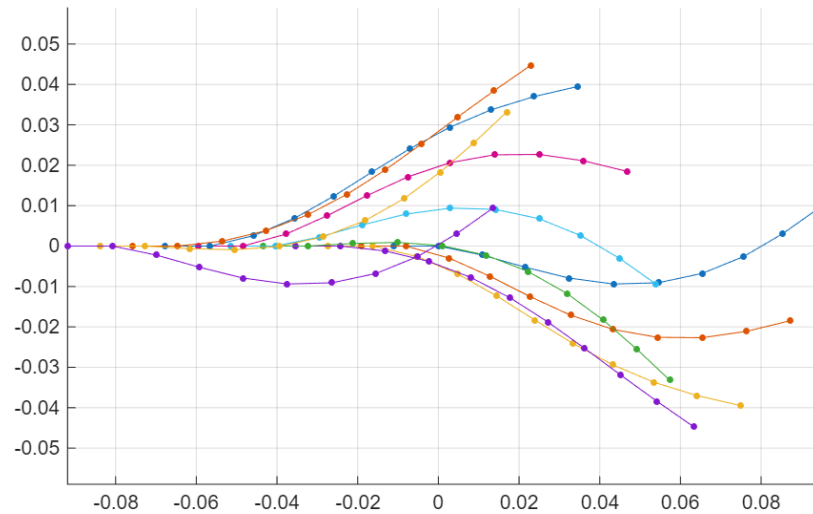
We also obtain an associated optimal control phasor, from which we can obtain the optimal control trajectory and optimal gait.



**Figure 1:** Optimal control signal and gait trajectory.



**Figure 2:** Optimal control and gait phasor properties.



**Figure 3:** Snapshots of optimal gait through one cycle.

(b)

Our overarching goal is to design a controller in the form of

$$\begin{aligned}\hat{\dot{x}} &= A\hat{x} + Bu + \mathcal{F}(C\hat{x} - y) \\ \dot{\xi} &= \psi(\xi) + G(C\hat{x} - y) \\ u &= U\xi + \mathcal{K}(\hat{x} - X\xi)\end{aligned}$$

where  $\sigma(r) = \mu(\alpha^2 - r^2)$  (i.e. the nonlinear expression from Homework 4). With the knowledge that we want a nonlinear output regulator, we can immediately set  $G = 0$ .

We can start the process by determining the stable limit cycles that we want to achieve and their appropriate representation for the design problem (i.e.  $R_k$ ,  $\Omega_k$ , and  $\eta$ ). We can declare that our target will be the gaits  $\phi_k(t) = \Im(\hat{\phi}^{(k)} e^{j\omega_k t})$  that are optimal for their respective cost metrics with  $k = 1$  to be associated with the power-rate optimal gait and  $k = 2$  to be associated with the shape-optimal gait.

Let us first figure out what our plant will be. With state vector  $x := \text{col}(\theta, \dot{\theta})$  defined for us, let us also assume that we are following along the nominal locomotor velocity — i.e. assume that  $\dot{w} = v_o$ . Our state dynamics become:

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & I \\ -J^{-1}(v_o\Lambda + k_o BB^T) & -J^{-1}D \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ J^{-1}F^T \end{bmatrix}}_{\mathcal{B}_1} f + \underbrace{\begin{bmatrix} 0 \\ J^{-1}B \end{bmatrix}}_{\mathcal{B}_2} u$$

Assuming that output  $y = \phi$  is available for the controller, we have the output equation:

$$y = \underbrace{\begin{bmatrix} B^T & 0 \end{bmatrix}}_{\mathcal{C}_2} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

Additionally, we want the locomotor to converge to an orbitally stable gait. From the S-procedure we have recovered optimal gait  $\phi_k$  and optimal control input  $u_k$ . Let us define our target then by stacking these two signals.

$$p = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} \phi \\ u \end{bmatrix} = \underbrace{\begin{bmatrix} B^T & 0 \\ 0 & 0 \end{bmatrix}}_{\mathcal{E}} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ I \end{bmatrix}}_{\mathcal{D}} u$$

Through the S-procedure, we procure a gait and control phasor  $(\hat{\phi}^{(k)}, \hat{u}^{(k)})$  and an optimal frequency  $\omega_k$  for each gait, wherein

$$\begin{bmatrix} \hat{\phi}_i^{(k)} \\ \hat{u}_i^{(k)} \end{bmatrix} = \begin{bmatrix} \alpha_i^{(k)} e^{j\varphi_i^{(k)}} \\ \beta_i^{(k)} e^{j\nu_i^{(k)}} \end{bmatrix}$$

From the procedure derived in the notes, we have that for each target gait  $\rho_k = [\phi_k(t) \ u_k(t)]^T = \Im([\hat{\phi}^{(k)} e^{j\omega_k t} \ \hat{u}^{(k)} e^{j\omega_k t}])^T$ ,

$$\Omega_k = \begin{bmatrix} 0 & -\omega_k \\ \omega_k & 0 \end{bmatrix}, \quad R_k = \begin{bmatrix} \vdots & \vdots \\ \alpha_i^{(k)} \sin \varphi_i^{(k)} & \alpha_i^{(k)} \cos \varphi_i^{(k)} \\ \vdots & \vdots \\ \beta_i^{(k)} \sin \nu_i^{(k)} & \beta_i^{(k)} \cos \nu_i^{(k)} \\ \vdots & \vdots \end{bmatrix}, \quad \eta = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

We can collate matrices by defining  $\Omega := \text{diag}(\Omega_1, \Omega_2)$  and  $R := \begin{bmatrix} R_1 & R_2 \end{bmatrix}$ . We obtain the following matrices:

we obtain the following matrices:

$$\Omega = \begin{bmatrix} \begin{bmatrix} 0 & -16.0246 \\ 16.0246 & 0 \end{bmatrix} & 0 \\ 0 & \begin{bmatrix} 0 & -11.6638 \\ 11.6638 & 0 \end{bmatrix} \end{bmatrix}$$

$$R = \begin{bmatrix} 0.1717 & -0.2371 & -0.1806 & 0.2022 \\ 0.2212 & 0.3569 & -0.1550 & 0.0725 \\ -0.2318 & 0.4268 & -0.1250 & -0.0320 \\ -0.5687 & 0.0605 & -0.0865 & -0.1099 \\ -0.4619 & -0.4751 & -0.0416 & -0.1600 \\ 0.0709 & -0.7438 & 0.0036 & -0.1819 \\ 0.6597 & -0.4849 & 0.0404 & -0.1760 \\ 0.8565 & 0.1675 & 0.0587 & -0.1434 \\ 0.4806 & 0.7682 & 0.0484 & -0.0851 \\ -0.0000 & 0.0002 & -0.0000 & 0.0014 \\ -0.0000 & 0.0002 & -0.0000 & 0.0011 \\ -0.0001 & 0.0002 & -0.0000 & 0.0009 \\ -0.0001 & 0.0001 & -0.0000 & 0.0007 \\ -0.0001 & 0.0001 & -0.0000 & 0.0005 \\ -0.0001 & 0.0000 & -0.0000 & 0.0003 \\ -0.0000 & 0.0000 & -0.0000 & 0.0002 \\ 0.0000 & 0.0000 & -0.0000 & 0.0001 \\ 0.0000 & 0.0000 & -0.0000 & 0.0000 \end{bmatrix}$$

The next step is to find the dynamics of the oscillator  $\dot{\xi} = \psi(\xi)$  and specify  $H_k$  from which  $\xi = h_k = H_k e^{\Omega_k t} \eta$  is orbitally stable. Our process will inevitably require that  $H \triangleq \begin{bmatrix} H_1 & \dots & H_k \end{bmatrix} \in \mathbb{R}^{r \times 4}$  is full column rank, where  $r$  is the dimension of  $\xi(t)$  and  $h_k(t)$ . We shall choose the simplest case, in which  $r = 4$ . Now we choose an oscillator. Let's choose an AHO,  $\dot{z} = M\varphi(z)$ . If we choose to define

$$\xi \triangleq \begin{bmatrix} \Re(z) \\ \Im(z) \end{bmatrix}$$

it must be the case that  $z \in \mathbb{C}^2$ . For this oscillator, we want that  $z = \zeta_k = e^{j(\omega_k t + \phi_k)}$ . The values of  $\omega_k$  arrive from the optimal gait calculations that we have done above and in Homework 3. That leaves us with  $\phi_k \in \mathbb{R}^2$  as a free design parameter for us to choose. Let's set

$$\phi_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \phi_2 = \begin{bmatrix} 0 \\ \frac{\pi}{2} \end{bmatrix}$$

For each  $\phi_k$  and  $\omega_k$ , we derive the  $M$  matrix for the AHO network. I shall declare that the power-optimal gait will be associated with  $k = 1$  and that the shape-optimal gait will be associated with  $k = 2$ . We then define matrices  $V$  and  $j\Omega$  by

$$V \triangleq \begin{bmatrix} e^{j\phi_1} & e^{j\phi_2} \end{bmatrix}, \quad j\Omega = \begin{bmatrix} j\omega_1 & 0 \\ 0 & j\omega_2 \end{bmatrix}$$

We use these to create matrix  $M = V(j\Omega)V^{-1}$ . Plugging in numeric values, we obtain:

$$M = \begin{bmatrix} 2.1804 + 13.8442i & -2.1804 + 2.1804i \\ 2.1804 + 2.1804i & -2.1804 + 13.8442i \end{bmatrix}$$

We also have  $\varphi(z) = (1 + j\mu(|z|^2 - 1))z$ , which operates on each element of  $z \in \mathbb{C}^2$ . We will choose  $\mu = 100$  as in Homework 3. Using these definitions, we finally define the dynamics for  $\xi$ :

$$\underbrace{\begin{bmatrix} \Re(\dot{z}) \\ \Im(\dot{z}) \end{bmatrix}}_{\dot{\xi}} = \underbrace{\begin{bmatrix} \Re(M\varphi(z)) \\ \Im(M\varphi(z)) \end{bmatrix}}_{\psi(\xi)}$$

From our choice in  $\phi_k$ , we can also define matrices  $H_k$  by

$$H_k = \begin{bmatrix} \cos \phi_k & -\sin \phi_k \\ \sin \phi_k & \cos \phi_k \end{bmatrix}$$

and conjoin them to generate  $H$ :

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Our next step is to use  $(R_k, \Omega_k, H)$  to generate  $(X, U)$  for our controller. For each  $(\Omega_k, R_k)$ , we have a regulator equation

$$\begin{aligned} \mathcal{A}X_k + \mathcal{B}_2U_k &= X_k\Omega_k \\ \mathcal{E}X_k + \mathcal{D}U_k &= R_k \end{aligned}$$

Working some things around, we can write  $U_k$  in terms of  $X_k$  substituting this into the top equation, we have a Sylvester equation that can be solved for  $X_k$ , which is then used to solve for  $U_k$ .

$$\begin{aligned} \mathcal{E}X_k + \mathcal{D}U_k &= R_k \\ U_k &= \mathcal{D}^+R_k - \mathcal{D}^+\mathcal{E}X_k \\ \mathcal{A}X_k + \mathcal{B}_2(\mathcal{D}^+R_k - \mathcal{D}^+\mathcal{E}X_k) &= X_k\Omega_k \\ (\mathcal{A} - \mathcal{B}_2\mathcal{D}^+\mathcal{E})X_k - X_k\Omega_k &= -\mathcal{B}_2\mathcal{D}^+R_k \end{aligned}$$

We then define the cumulative  $X$  and  $U$  matrices by:

$$\begin{aligned} X &\triangleq \begin{bmatrix} \dots & X_k & \dots \end{bmatrix} H^+ \\ U &\triangleq \begin{bmatrix} \dots & U_k & \dots \end{bmatrix} H^+ \end{aligned}$$

Using this process, we get:

$$X = \begin{bmatrix} -0.2152 & 0.0435 & -0.1587 & 0.3958 \\ -0.1061 & -0.2868 & -0.5614 & 0.4417 \\ 0.3016 & -0.4627 & -0.7054 & 0.1589 \\ 0.7145 & -0.3069 & -0.4396 & -0.1674 \\ 0.8087 & 0.0608 & 0.0881 & -0.2200 \\ 0.4905 & 0.3082 & 0.5173 & 0.0946 \\ -0.0140 & 0.1530 & 0.5381 & 0.5587 \\ -0.3161 & -0.4013 & 0.1272 & 0.8021 \\ -0.1539 & -1.0442 & -0.4305 & 0.5916 \\ -1.7084 & 5.5085 & 3.4018 & -0.6500 \\ -7.6660 & 5.7477 & 1.8334 & 4.4623 \\ -9.7703 & 1.0132 & -4.3585 & 6.9402 \\ -5.5621 & -4.1653 & -10.5462 & 4.0139 \\ 2.6363 & -4.7497 & -11.6162 & -2.3181 \\ 9.1099 & 0.6964 & -6.1285 & -6.6703 \\ 8.9716 & 8.6053 & 2.2514 & -4.4795 \\ 1.9460 & 12.9464 & 7.2772 & 4.2191 \\ -7.2819 & 9.8639 & 4.7587 & 14.4398 \end{bmatrix}$$

$$U = 10^{-3} \cdot \begin{bmatrix} 0.5813 & -0.5813 & 0.7929 & -0.5813 \\ 0.4472 & -0.4611 & 0.6803 & -0.4593 \\ 0.3209 & -0.3738 & 0.5415 & -0.3439 \\ 0.2220 & -0.3041 & 0.3978 & -0.2527 \\ 0.1558 & -0.2373 & 0.2696 & -0.1896 \\ 0.1127 & -0.1632 & 0.1727 & -0.1440 \\ 0.0765 & -0.0853 & 0.1090 & -0.100 \\ 0.0375 & -0.0193 & 0.0681 & -0.0501 \\ -0.0010 & 0.0169 & 0.0376 & -0.0018 \end{bmatrix}$$

Next to do for the controller is to derive gains  $\mathcal{K}$  and  $\mathcal{F}$  for the controller<sup>1</sup>. We will require that  $\mathcal{A} + \mathcal{B}_2\mathcal{K} \in \mathbb{H}$  and  $\mathcal{A} + \mathcal{F}\mathcal{C}_2 \in \mathbb{H}$ . From the  $H_2$ -optimal control perspective, having no sensor noise creates issues for numeric calculations of the Kalman filter gain. As such, something we can do use a different design plant considers sensor noise. Let's say that each output channel receives its own noisy signal. We can thus use a design output:

$$y = \underbrace{\begin{bmatrix} B^T & 0 \\ 0 & 0 \end{bmatrix}}_{\mathcal{C}_2} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + 0.001 \cdot \underbrace{I}_{\mathcal{D}_{21}} \cdot v$$

We can derive an optimal Kalman filter gain for this design plant, where it will be the stabilizing solution to the Riccati equation:

$$\mathcal{F} = -\mathcal{Q}\mathcal{C}_2^T, \mathcal{A}\mathcal{Q} + \mathcal{Q}\mathcal{A}^T - \mathcal{Q}\mathcal{C}_2^T(\mathcal{D}_{21}\mathcal{D}_{21}^T)\mathcal{C}_2\mathcal{Q} + \mathcal{B}_1\mathcal{B}_1^T = 0$$

Doing so, we get the following Kalman gain:

<sup>1</sup>  $\mathcal{K}$  and  $\mathcal{F}$  are in a special font to differentiate them from system parameters  $F$  and  $K = k_o B B^T$ .

$$\mathcal{F} = 10^6 \cdot \begin{bmatrix} 0.0025 & -0.0018 & -0.0003 & 0.0000 & -0.0000 & 0.0000 & -0.0000 & 0.0000 & -0.0000 \\ 0.0007 & 0.0028 & -0.0020 & -0.0006 & 0.0000 & -0.0001 & 0.0001 & -0.0000 & 0.0001 \\ 0.0003 & 0.0012 & 0.0026 & -0.0019 & -0.0007 & 0.0001 & -0.0002 & 0.0001 & -0.0001 \\ 0.0003 & 0.0005 & 0.0014 & 0.0026 & -0.0019 & -0.0007 & 0.0001 & -0.0002 & 0.0002 \\ 0.0003 & 0.0006 & 0.0006 & 0.0015 & 0.0027 & -0.0018 & -0.0007 & 0.0001 & -0.0002 \\ 0.0003 & 0.0004 & 0.0008 & 0.0007 & 0.0016 & 0.0027 & -0.0017 & -0.0006 & 0.0004 \\ 0.0003 & 0.0005 & 0.0005 & 0.0009 & 0.0008 & 0.0018 & 0.0029 & -0.0015 & -0.0006 \\ 0.0003 & 0.0005 & 0.0007 & 0.0007 & 0.0011 & 0.0010 & 0.0020 & 0.0033 & -0.0008 \\ 0.0003 & 0.0005 & 0.0005 & 0.0009 & 0.0007 & 0.0016 & 0.0011 & 0.0031 & 0.0037 \\ 4.7911 & -6.2007 & 0.2833 & 0.9035 & -0.1448 & 0.2646 & -0.0381 & 0.0775 & -0.0439 \\ -1.4087 & 7.6303 & -6.4766 & -1.0409 & 1.3251 & -0.4640 & 0.4302 & -0.1504 & 0.2114 \\ -1.1262 & 0.8728 & 6.6333 & -6.1072 & -1.4330 & 1.4452 & -0.5997 & 0.4696 & -0.3423 \\ -0.2223 & -1.0729 & 1.5694 & 6.3442 & -5.9843 & -1.5743 & 1.5218 & -0.6763 & 0.7075 \\ -0.3675 & 0.3975 & -1.1896 & 1.7951 & 6.2900 & -5.8887 & -1.5935 & 1.5547 & -1.0066 \\ -0.1026 & -0.3315 & 0.7200 & -1.2253 & 1.9779 & 6.2543 & -5.6645 & -1.6536 & 2.1977 \\ -0.1410 & 0.1372 & -0.3105 & 0.8969 & -1.1389 & 2.1862 & 6.4391 & -5.3361 & -2.2328 \\ -0.0633 & -0.0905 & 0.3090 & -0.2480 & 1.0911 & -1.0210 & 2.7561 & 6.7426 & -2.9803 \\ -0.1077 & 0.1660 & -0.2471 & 0.8053 & -0.6282 & 2.1903 & -1.6835 & 6.0127 & 7.8361 \end{bmatrix}$$

We know that this is a feasible solution because the eigenvalues of  $\mathcal{A} + \mathcal{F}\mathcal{C}_2$  have negative real eigenvalues:

$$\text{eig}(\mathcal{A} + \mathcal{F}\mathcal{C}_2) = 10^3 \cdot \begin{bmatrix} -3.4999 + 3.5107i \\ -3.4999 - 3.5107i \\ -3.3084 + 3.3179i \\ -3.3084 - 3.3179i \\ -3.1226 + 3.1352i \\ -3.1226 - 3.1352i \\ -2.8039 + 2.8094i \\ -2.8039 - 2.8094i \\ -0.3995 + 0.3947i \\ -0.3995 - 0.3947i \\ -0.9849 + 0.9832i \\ -0.9849 - 0.9832i \\ -2.3170 + 2.3238i \\ -2.3170 - 2.3238i \\ -1.5961 + 1.5985i \\ -1.5961 - 1.5985i \\ -1.9299 + 1.9518i \\ -1.9299 - 1.9518i \end{bmatrix}$$

As a result, the state estimate will converge to the true state. The next thing we should take care of is the state feedback gain. We can fall back into  $H_2$ -optimal control theory and let this gain be the LQR gain from solving for the stabilizing solution of the following Riccati equation:

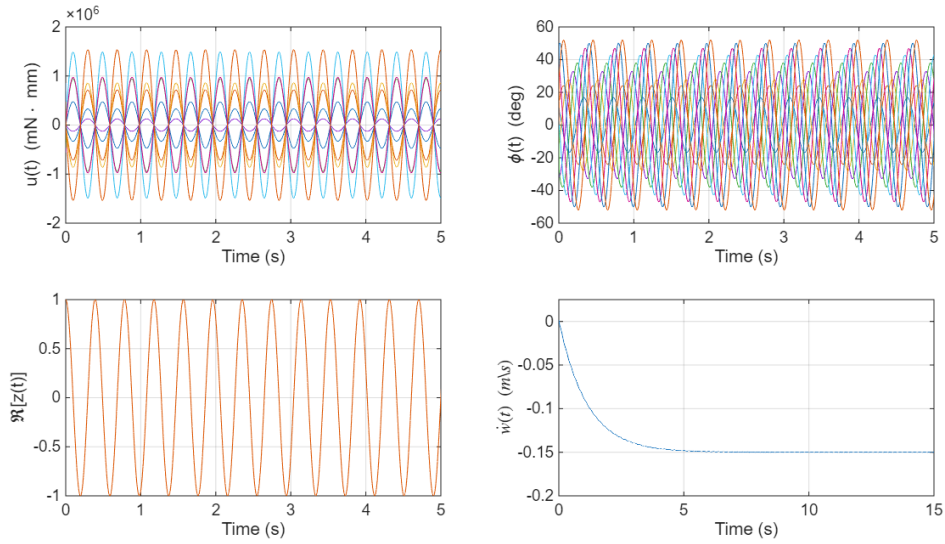
$$\mathcal{K} = -\mathcal{B}_2^T P, \quad P\mathcal{A} + \mathcal{A}^T P - P\mathcal{B}_2(\mathcal{D}^T \mathcal{D})^{-1} \mathcal{B}_2^T P + \mathcal{E}^T \mathcal{E} = 0$$

We get a numeric value for the LQR gain, but I will not present it for reasons I will discuss later.

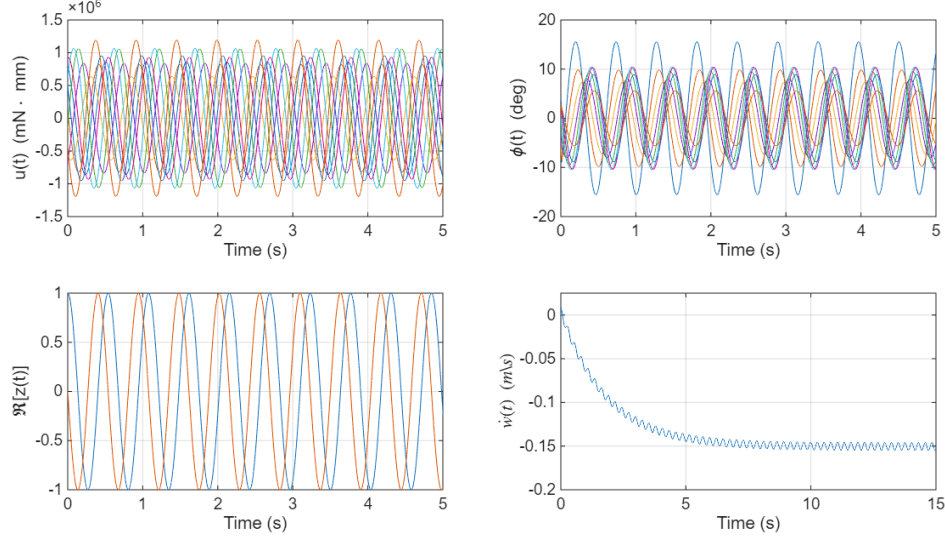
We have now constructed all relevant matrices for our nonlinear output regulator. Now we can begin the simulation aspect of the problem. We want to extract the two different gaits from zero state initial conditions, with the only difference being from the complex oscillator state  $z(0)$ . Working with the  $\xi$  representation, we know that it will converge to target  $h_k(t) = H_k e^{\Omega_k t} \eta$ . Though we do not know that domain of attraction analytically, we know that if we start on this trajectory we will remain on this trajectory. Thus, we can start our oscillator on  $\xi(0) = h_k(0) = H_k e^{\Omega_k \cdot (0)} \eta$  and we will guarantee that we will remain on that oscillation. To move into our  $z$  coordinates, we follow the definition  $z \triangleq \xi_1 + j\xi_2$ , where  $\xi_1$  is the top half of  $\xi$  and  $\xi_2$  the bottom. Performing this, we have that our initial oscillator states should be

$$z_{power}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad z_{shape}(0) = \begin{bmatrix} 1 \\ j \end{bmatrix}$$

These will prove to be sufficient initial conditions for our simulations.



**Figure 4:** Simulation results for power-optimal gait.  $z(0) = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ .



**Figure 5:** Simulation results for shape-optimal gait.  $z(0) = \begin{bmatrix} 1 & j \end{bmatrix}^T$ .

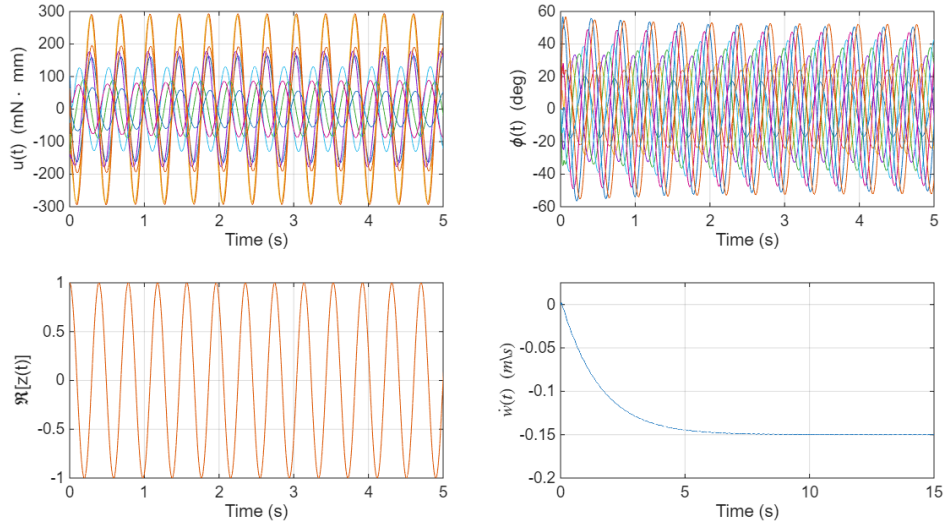
So I am actually quite disturbed by these results, specifically just the signal  $u(t)$  for both gaits. They are significantly larger in magnitude than the target we wanted them to converge to: a  $10^4$  magnitude difference for the power-optimal gait and an  $10^3$  magnitude difference for the shape-optimal gait. This signifies to me that the the state feedback gain that we obtained from the  $H_2$ -optimal control scheme is far too strong to be effective in our controller. What we can do is scale  $\mathcal{K}$  down until our simulated  $u(t)$  signals look appropriate and match the targets we want them to achieve. After some tuning, we get decent results by multiplying our original LQR gain by 0.0001:

$$\mathcal{K} = 10^{-3} \cdot \begin{bmatrix} 0.1000 & -0.1000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ -0.0000 & 0.1000 & -0.1000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ -0.0000 & -0.0000 & 0.1000 & -0.1000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ -0.0000 & -0.0000 & -0.0000 & 0.1000 & -0.1000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ -0.0000 & -0.0000 & -0.0000 & -0.0000 & 0.1000 & -0.1000 & -0.0000 & -0.0000 & -0.0000 \\ -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & 0.1000 & -0.1000 & -0.0000 & -0.0000 \\ -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & 0.1000 & -0.1000 & -0.0000 \\ -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & 0.1000 & -0.1000 \\ -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & 0.1000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 & -0.0000 & -0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 & -0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \end{bmatrix}^T$$

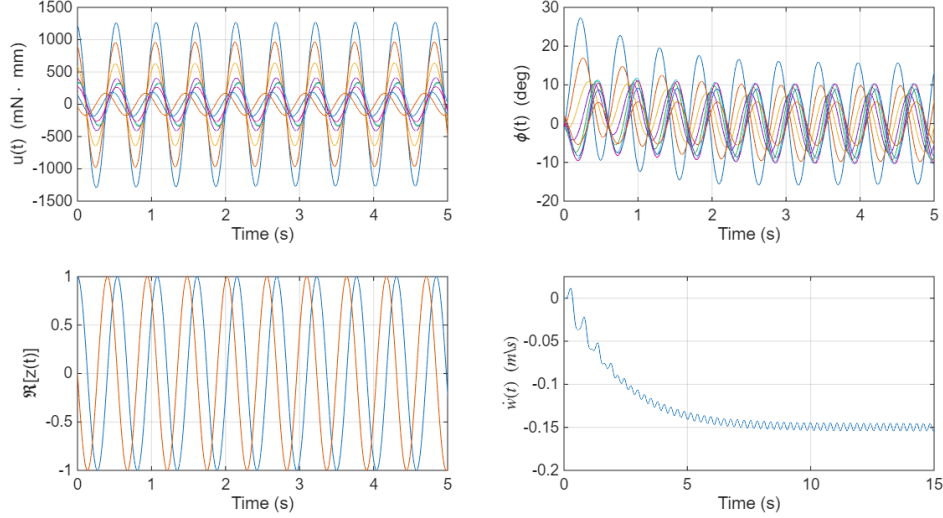
We can verify that this is a feasible choice in  $\mathcal{K}$  by determining the eigenvalues of the closed-loop system:

$$\text{eig}(\mathcal{A} + \mathcal{B}_2\mathcal{K}) = 10^2 \cdot \begin{bmatrix} -0.4899 + 6.8227i \\ -0.4899 - 6.8227i \\ -0.4847 + 5.9307i \\ -0.4847 - 5.9307i \\ -0.4776 + 4.7138i \\ -0.4776 - 4.7138i \\ -0.4702 + 3.4500i \\ -0.4702 - 3.4500i \\ -0.4636 + 2.3107i \\ -0.4636 - 2.3107i \\ -0.4582 + 1.3628i \\ -0.4582 - 1.3628i \\ -0.4542 + 0.5990i \\ -0.4542 - 0.5990i \\ -0.0391 + 0.0000i \\ -0.1227 + 0.0000i \\ -0.7805 + 0.0000i \\ -0.8610 + 0.0000i \end{bmatrix}$$

All of the eigenvalues of  $\mathcal{A} + \mathcal{B}_2\mathcal{K}$  have negative real eigenvalues, so this is a suitable choice in LQR gain. Simulating the two gaits with this new LQR gain, we obtain the following results:



**Figure 6:** Simulation results for power-optimal gait, new state feedback gain.



**Figure 7:** Simulation results for shape-optimal gait, new state feedback gain.

I feel that these results are much better. The magnitude of  $u(t)$  are now comparable to the prior target control oscillations we established before. This signal kicks into oscillation immediately because the small  $\mathcal{K}$  means that it is mainly determined by  $U\xi$ . It appears that  $\Re[z(t)]$  for both cases experience no transients. This result is by design. We start  $z(t)$  exactly on the stable periodic orbit, so it naturally follows the oscillation immediately. It is also cool to notice that we see the underlying choice of phases for  $H_k$ . We can observe that  $\Re[z(t)]$  for the power-optimal gait is synchronized, corresponding to the synchronized phases we assigned to it, and that for the shape-optimal gait we see the quarter-phase delay that we prescribed.

We see that because the physical states of the flipper start from zero, they take some time to enter their periodic orbits. The transient time constant is about 0.6 seconds for the power-optimal gait and about 1.5 for the shape-optimal gait. I would say that this value comes from the combined effects of state estimation and state feedback. Each of these are associated with a stable closed-loop system matrix ( $\mathcal{A} + \mathcal{F}\mathcal{C}_2$  and  $\mathcal{A} + \mathcal{B}_2\mathcal{K}$ , respectively) which have different eigenvalues for each associated mode/oscillation. The decay rate, and thus time constant, for the transient will arrive the decay rates of the modes that are the most prominent based on the phase profile of each gait.

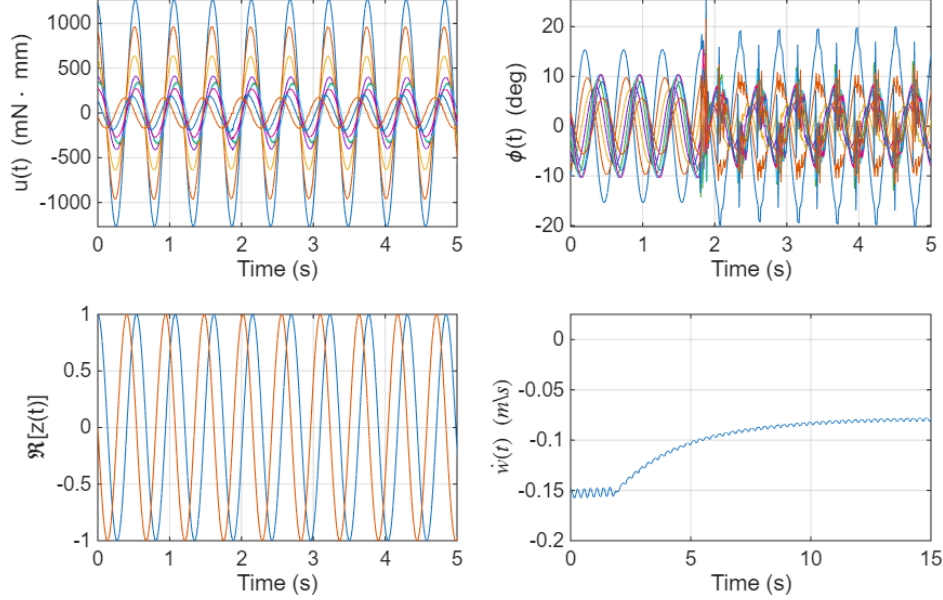
It would appear that the time constant for  $\dot{w}$  is 1.8 seconds for the power-optimal gait and 2.5 seconds for the shape-optimal gait. I think that this originates from the dynamics of linear motion for the locomotor. We see that the rate of decay for  $\dot{w}$  is dictated by  $d(\theta)/m$ . For the shape-optimal gait, the values of  $\theta$  are generally low (low curvature from link to link), so this parameter is larger and will produce a larger time constant. As for the power-optimal gait, its values of  $\theta$  are larger (high curvature from link to link), thus  $d(\theta)$  is larger and it has a shorter time constant.

### (c)

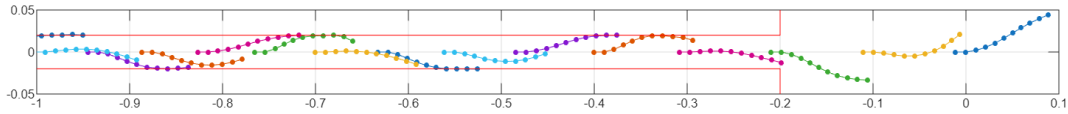
We should base the initial state based on what we would like to achieve. First and foremost we are provided that  $w(0) = 0$ , so that must be included in our initial state. Next, we want our target velocity to be  $v_o = -0.15$  m/s, so we will set  $\dot{w}(0) = v_o = -0.15$ . From the previous section, we have the form of the initial complex state of the oscillator  $z(0)$ . We will use the same form for the respective gaits. We know that our estimated state will eventually converge to the true state due to the convergence of the error to zero ( $\mathcal{A} + \mathcal{F}\mathcal{C}_2 \in \mathbb{H}$ ), so we can set them arbitrarily to zero. That brings us to the initial true states  $\theta(0)$  and  $\dot{\theta}(0)$ . If we want to initialize the simulation on one of our optimal gaits, we should have that the initial state be along the periodic orbit of said optimal gait. Combining this with phasor notation  $\theta(t) = \Im[\hat{\theta}_k e^{j\omega_k t}]$ , a valid initial angle vector will be when  $t = 0$ . Thus we obtain  $\theta(0) = \Im[\hat{\theta}_k]$  for our angular position. Taking a derivative of the phasor notation, we have that  $\dot{\theta}(t) = \Im[j\omega_k \hat{\theta}_k e^{j\omega_k t}]$ . Using a similar reasoning, we can take  $\dot{\theta}(0) = \Im[j\omega_k \hat{\theta}_k]$  to be our initial velocity vector. To summarize:

$$\begin{bmatrix} w(0) \\ \dot{w}(0) \end{bmatrix} = \begin{bmatrix} 0 \\ v_o \end{bmatrix}, \quad z(0) = H_k \eta[1:2] + j H_k \eta[3:4], \quad \begin{bmatrix} \hat{\theta}(0) \\ \dot{\hat{\theta}}(0) \end{bmatrix} = \mathbf{0}^{18}, \quad \begin{bmatrix} \theta(0) \\ \dot{\theta}(0) \end{bmatrix} = \begin{bmatrix} \Im[\hat{\theta}_k] \\ \Im[j\omega_k \hat{\theta}_k] \end{bmatrix}$$

Simulating from this initial condition with the environment boundaries, we obtain the following:



**Figure 8:** Simulation of flipper in environment starting from shape-optimal gait.



**Figure 9:** Snapshots of flipper position in environment.

We do not see any initial transient before convergence to the shape-optimal gait. This is because we have designed all initial state parameters (excluding state estimate) to be near or exactly on their optimal trajectories. Because they are so close to the stable periodic orbit, it takes an imperceptible amount of time to converge to the stable solution. When the tail strikes the walls, the gaits of the larger-amplitude are interrupted and exhibit sporadic behavior until the move away from the wall. This is caused by the disturbance reaction force that forces the link angle away from the wall while, at the same time, the control input continues to drive it into the wall. As a result, there is an incredibly high frequency scratching/oscillation whenever the tail strikes the wall. In this condition, locomotion speed  $v_o$  cannot be maintained in the passage, as seen in Figure 8. Constant  $v_o$  was only achievable when the optimal gait was maintained. Because, it is be disturbed by the wall force, though, we do not expect to see  $\dot{w}(t) \approx v_o$  through the passage.

We can also notice that the complex oscillatory state is not affected by the tail striking the wall, nor is the control signal which is largely reliant on the exact same complex state.

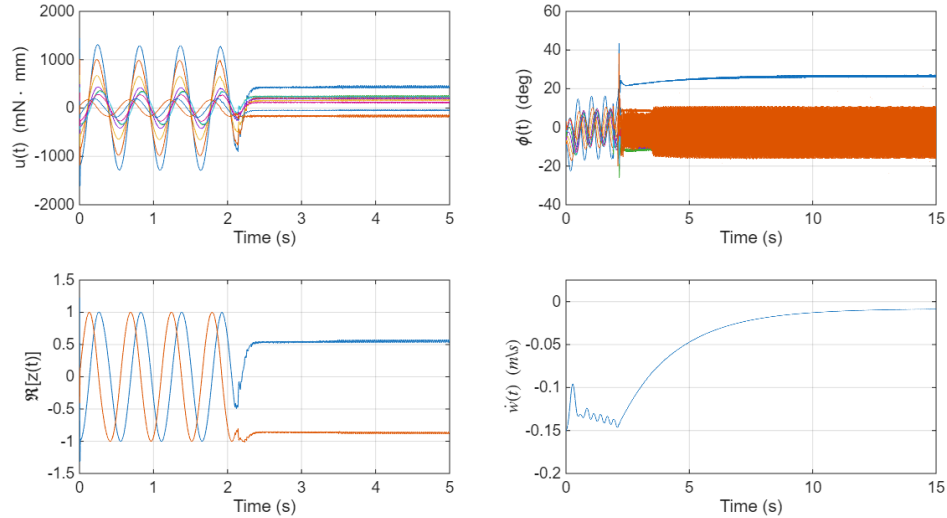
I would say that the control design is successful in going through the narrow passage, but in its current state is not operating to the full potential of its dynamics. I obtain my notion of success based on the fact that it retains a non-zero average velocity once it enters the passage. Although it is not the target/design velocity  $v_o$ , it would still be able to traverse through to the other end of the passageway. However, we have proved that the complex oscillator system that we designed is capable of entering another gait — one that has smaller amplitude gait oscillations. If we were to enter this gait, we would be able to maintain our desired velocity  $v_o$  once we entered the passage. However, based on how we designed the controller (specifically the choice

of an output regulator instead of an autonomous pattern formation), we cannot access the shape-optimal gait. Because the complex oscillator dynamics do not consider the state of the flipper, it is essentially locked into its initialized gait. In this sense, the control design fails to make the best possible use of the designed dynamics.

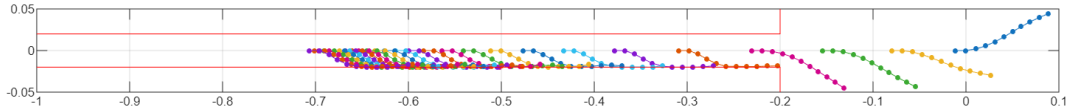
(d)

We can correct our design by taking a hint from the Autonomous Pattern Formation we explored in Homework 4. What this means is that we utilize a sensory feedback term  $G(\mathcal{C}_2\hat{x} - y)$  to the dynamics of  $\xi$ , which allows for deviation in the oscillation behavior. What we hope to achieve is that the discrepancy between the predicted and true output will push the oscillation state from the initial shape-optimal periodic orbit into the domain of attraction of the power-optimal periodic orbit. The output regulator formulation required that  $G = 0$ . We will be changing its value to generate the adaptive behavior we want to see.

The challenge will be in designing this parameter. My gut instinct when first seeing this was to turn the dual regulator equation, since we have used this previously in Homework 4. However, when I attempted this the controller utter failed. As shown in Figures 10 and 11, it seems like the flipper hits the wall with its tail and just gives up.



**Figure 10:** Autonomous Pattern Formation controller with dual regulator-derived sensory feedback gain.



**Figure 11:** Snapshots of APF controller with dual regulator-derived sensory feedback gain.

This behavior seems really weird. My proposition for what occurred is that that this  $G$  introduces a new equilibrium point for the oscillator dynamics. When the oscillator state is perturbed by the sensory feedback, the oscillator is attracted to this new equilibrium instead of the power-optimal limit cycle. Figure 10 seems to support this behavior, as the oscillator states remain constant after a certain point after hitting the wall. Perhaps that is because we framed this definition of  $G$  to be optimal in  $H_2$ , where we never necessarily judged its performance against non-impulse disturbances.

So how do we fix this? My next intuition would be from section (b). We designed the oscillator trajectories with underlying phase vectors

$$\phi_{power} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \phi_{shape} = \begin{bmatrix} 0 \\ \frac{\pi}{2} \end{bmatrix}$$

We note that the difference between them is a phase difference of 90 degrees in the second element. Thus, to make the shape-optimal oscillation to converge to the power-optimal oscillation, we need to rotate  $z_2$  by 90 degrees in the complex plane. This is more easily illustrated when we see the initial complex states for the respective optimal gaits.

$$z_{power}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad z_{shape}(0) = \begin{bmatrix} 1 \\ j \end{bmatrix}$$

With the first element the same, we can see that the second element  $z^{(2)}$  of the shape-optimal gait can be derived from the second of the power-optimal gait by a complex-plane rotation:

$$\begin{bmatrix} \Re[z_{power}^{(2)}] \\ \Im[z_{power}^{(2)}] \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} \Re[z_{shape}^{(2)}] \\ \Im[z_{shape}^{(2)}] \end{bmatrix}$$

This is a neat thing to notice, but we need to somehow connect this to the dynamics of  $\xi$ . We see that, overall, to get to the power gait, the real part of the shape gait is made negative and the imaginary part of the shape gait is more positive<sup>2</sup>. Thus, when we construct the gain matrix  $G$ , it should react to sensory feedback negative in the channel related to the real component of  $z^{(2)}$  and positively in the channel related to the imaginary component of  $z^{(2)}$ .

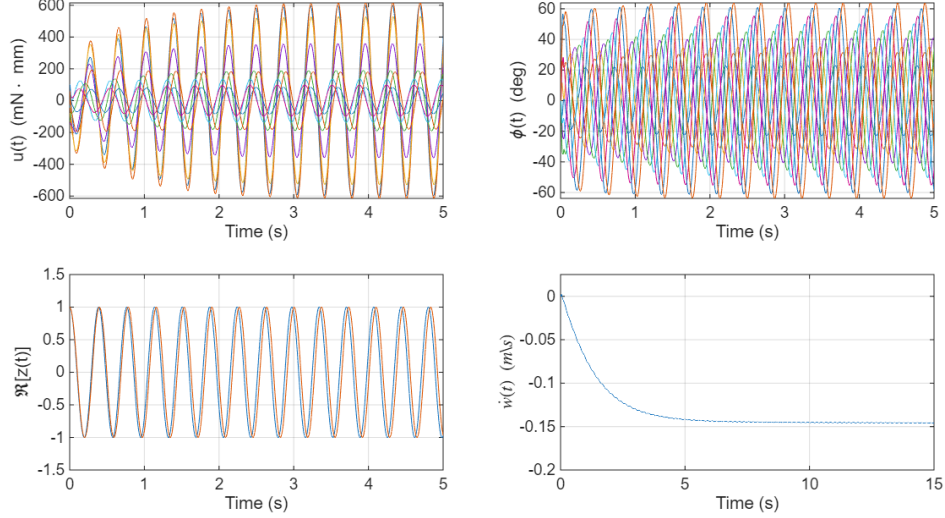
$$G = \begin{bmatrix} \dots & 0 & \dots \\ \dots & - & \dots \\ \dots & 0 & \dots \\ \dots & + & \dots \end{bmatrix}$$

Since we want the oscillator to react to discrepancies in all joints, all elements in the nonzero rows should be occupied. I will choose them to all be equal, but I am sure you could choose them to prioritize the effects of errors at the head versus errors at the tail. Finally, we must choose the magnitude at which to apply them. Because the "rotation" is applied equally on the real and imaginary channels, **every** nonzero element will be the same magnitude. I discovered the magnitude by trial and error, and discovered the following gain matrix to perform the best:

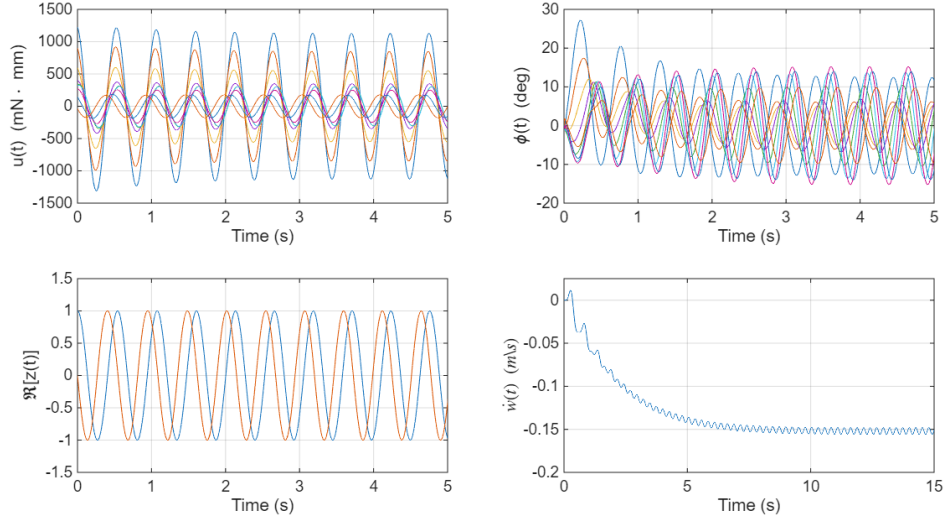
$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1500 & -1500 & -1500 & -1500 & -1500 & -1500 & -1500 & -1500 & -1500 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1500 & 1500 & 1500 & 1500 & 1500 & 1500 & 1500 & 1500 & 1500 \end{bmatrix}$$

We can now simulate the system and compare it to the performance under the previous controller.

<sup>2</sup> Strictly speaking, I believe I would need to get to the dynamics equations to make a full connection, but I feel like this would be good enough for the purposes of this assignment.



**Figure 12:** Power-optimal gait of APF controller with updated sensory feedback gain.

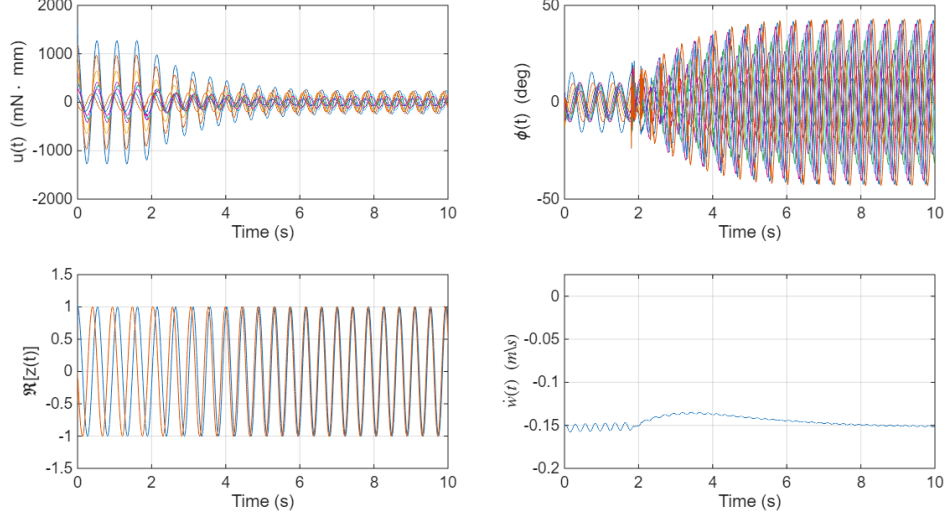


**Figure 13:** Shape-optimal gait APF controller with updated sensory feedback gain.

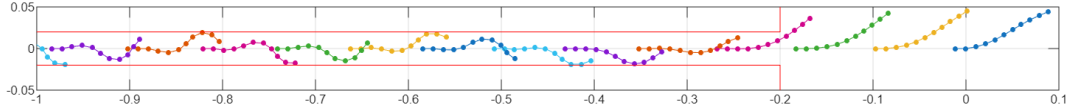
This is great! This new controller successfully meets specification (S1). Comparing to the controller in part (b), Figures 12 and 13 show some nuances that are introduced to this controller. The performance of  $\phi(t)$  appear to be near identical from (b) to (d), but there does appear to be a slight degradation in the performance of  $\dot{w}(t)$  in the power-optimal case in the new controller. For the root of this cause, we can turn to the plots of  $\Re[z(t)]$  and  $u(t)$ .

I believe that a core difference with this new controller is the difference in convergence of the complex oscillator state. In part (b), we see that the complex oscillator state begins on its orbitally stable solution and remains on it for all time. Moreover, this solution is based solely on the design parameters  $\phi_k$  that were used to design  $M$ . However, in the case of our new controller, the orbitally stable solution is actually slightly different than before, with slight alterations in phase for both the power- and shape-optimal gaits. Because we start from an initial state that is not along this solution, there is some slight transient behavior. We can see the effects of this clearly in the plots of  $u(t)$ , which now have transient behavior as well. Because the convergence of the complex oscillator is not immediate, there is also some time delay for the control signal to converge as well.  $u(t)$  actually begins exactly as it did before in part (b), with the same initial amplitude. However, it appears that the new solution to the oscillator affects what the steady state solution

for the control will be. In the case of the power-optimal gait, the amplitude is nearly doubled while in the shape-optimal case it is decreased very slightly. As a result of this deviation in control input, the system's states don't actually reach the true optimal gaits. Thus, the performance of  $\dot{w}(t)$  degraded, since the system isn't behaving optimally. That is why we see a slight deviation in convergence for the shape-optimal gait and a more obvious deviation in steady state velocity in the power-optimal gait.



**Figure 14:** Passage navigation of APF controller with updated sensory feedback gain.



**Figure 15:** Passage navigation snapshots of APF controller with updated sensory feedback gain.

The above plots are also really good to see and indicate a successful controller! We see that the flipper is initialized into the shape-optimal gait, as desired, and then strikes the wall 2 seconds into the simulation. Once this occurs, the effects of this disturbance are fed back into the oscillator, which gradually changes from the shape-optimal solution to its power-optimal solution. This causes the control input to converge to its power-optimal solution, which in turn does the same for the gait. The adaptation of the new gait is decently quick, with convergence to the target velocity occurring in around 6 seconds after initial impact with the walls. It is also cool to notice that the plot of  $\dot{w}(t)$  is implicitly telling us what gait it is using, as we know that the low and high frequency oscillations are associated with the shape- and power-optimal gaits, respectively. Figure 15 also confirms that specification (S2) is met. The flipper is very obviously in the shape optimal gait that does not fit in the walls. It is only *after* the flipper hits the walls that the switch in gait happens. Moreover, the relatively even spacing of the snapshots indicate that the flipper does not experience significant deceleration and maintains a fairly constant velocity.

While determining the best value of  $G$ , I experimented with the magnitude of its entries and with  $\mu$  as well. With my experiments with  $\mu$ , I was aware that it would effect the size of the domain of attraction for each complex oscillator limit cycle. Using low values would trap the flipper into its initial gait, meaning it would almost never transition into another gait. Increasing it makes the transition happen more easily, but the effects diminish after  $\mu = 100$ . Thus, that is why I stuck to this value. For the entries of  $G$ , increasing it generally means better performance for passage navigation, with shorter transient time after initial contact with the wall. However, as this value increases, we see more error in the steady state value of  $\dot{w}(t)$  as seen in Figure 12. In fact, when entries are around 2500, the average velocity actually approach  $v_o$  but diverge over time. Additionally, after entries were beyond 3000, the shape-optimal gait began to resemble the power-optimal gait, with actually very little difference in gait amplitude and oscillation frequency. I assume that this

is due to the fact that the high gain value would make the oscillator extremely sensitive to sensor feedback and will launch itself to the power-optimal gait.